

Chapter 175

Editing Basics in the Fairlight Page

You can use the Fairlight page to refine the editing of audio that was initially assembled in the Edit page, or you can use the Fairlight page to both record and edit audio programs from scratch.

Because audio clips have properties that video clips do not, audio editing encompasses additional procedures that are not available in the Edit page. This chapter takes you through the fundamental steps of editing audio the Fairlight way.

Contents

Compatible Audio Formats	3774
Editing Audio Clips Into the Timeline	3774
Overwriting Vs. Layering Clips That Overlap	3775
Choosing Parts of Clips to Edit in the Media Pool	3775
Dragging Audio Clips Into the Timeline	3776
Moving Audio Clips to Embedded Timecode Position	3778
Support for Mixed Audio Track Formats from Source Clips	3779
Making Audio Clip Selections in the Timeline	3780
Fairlight Edit Mode Remains Between Application Restarts	3780
Selecting Tracks	3780
Using Pointer Mode	3782
Using Range Mode	3783
Using Focus Mode	3788
Commands For Editing and Extending the Selection	3790
Locking Audio Tracks	3791
Splitting Clips	3791
Remove Silence (Studio Version Only)	3791
AI Checkerboard to New Tracks (Studio Version Only)	3792
Manually Split Clips in Pointer or Range Mode	3793
Linked Clips in the Fairlight Page	3794

Trimming Clips Without Rippling the Timeline	3794
Multi-Point Editing Overview	3794
Resizing the In and Out Points of a Clip	3795
Trim Start and Trim End	3795
Trim to Selection	3795
Moving and Overwriting Clips	3796
Snap to Playhead	3797
Subframe Nudging	3797
Slipping	3797
Duplicating Clips	3798
Disabling and Re-Enabling Clips in the Timeline	3799
Deleting Audio Clips and Regions	3799
Cut, Copy, and Paste	3799
Conventional Cut, Copy, and Paste	3799
Using the Cut/Copy Head and Tail Commands	3802
Paste and Remove Attributes	3803
Clip Attributes Naming	3803
Copying and Pasting Clip Attributes	3803
Copying and Pasting Track Attributes	3803
Removing Attributes	3804
EQ, Level, and AI Dialogue Matching	3804
EQ Matcher	3804
Level Matcher	3805
AI Dialogue Matcher	3806
Audio Clip Layering	3806
Audio Compound Clips	3808
Audio Crossfades	3808
Fades and Crossfades	3809
Using Fades	3809
Batch Fade and Crossfade Editor in the Fairlight Page	3810
Fade In and Out to Playhead	3812
Creating Crossfades With Overlapping Fades	3813
Using Crossfades From the Edit Page	3814
Finding Clips in the Media Pool	3814
Changing Clip Color in the Timeline	3814
Editing Audio Clips in External Editors	3815
Exporting Audio Clips to External Files	3816
Sample Editing	3817
Waveform Zero Crossing Indicator	3818
Voice Convert (Studio Version Only)	3818
Using Voice Convert	3819
Voice Convert Settings	3820
Creating Your Own Voice Models	3821

Compatible Audio Formats

DaVinci Resolve is compatible with WAVE, Broadcast WAVE, AIFF, MP3, AAC (M4A), CAF (macOS only), both MTS and QuickTime containers that use the AC3 audio format, and Enhanced AC-3 (macOS and Windows only). DaVinci Resolve is compatible with audio at sample rates including 32, 44.1, 48, 88.2, 96, and 192 kHz. Linux users have the added ability to encode and decode MP3 files in Fairlight.

DaVinci Resolve 17 adds Dolby Atmos ADM file creation and manipulation as well as the ability to export Dolby Atmos masters as IMF deliverables. IMF (Interoperable Master Format) is a SMPTE standard for a single master file format that incorporates all the media and metadata necessary to deliver what's necessary.

ADM Import

Importing an ADM file into a project will open the Bed mix or mixes and any associated Object tracks into a timeline. They will be imported onto corresponding track types of the audio files embedded into the ADM. For instance, if a Bed file is 7.1.2, then on import that file will open in a 7.1.2 track in Fairlight. Object files will be created as separate tracks in Fairlight containing all of their panning data.

NOTE: If bringing in a Dolby Atmos file from the Media Pool, it will render the file dynamically to the chosen output format directly on the Timeline using the Dolby Renderer. However, Fairlight can import the full Atmos file with beds and objects if imported from the Fairlight menu instead - Fairlight > Immersive > Audio > Import Master File. By bringing a file in this way, it will create a timeline with all of that file's Atmos program and dynamic metadata properly mapped and routed.

Editing Audio Clips Into the Timeline

The Fairlight page offers a complete audio editing environment that lets you either record and assemble clips from scratch, or refine tracks full of audio clips that have been edited together in different ways. There are four ways of adding media to the Timeline in the Fairlight page, depending on the type of work you do:

- Recording new audio into one or more tracks,.

For more information, see *Chapter 173, "Recording."*

- By dragging and dropping new audio clips from the Media Pool into the Fairlight timeline
- By editing audio clips into audio tracks on the Edit page
- By importing a project with audio clips
- By auditioning and confirming sound effects from the Sound Library

However audio clips come to be in your timeline, the rest of this chapter covers the myriad methods available to edit and sweeten the contents.

Overwriting Vs. Layering Clips That Overlap

When you add clips to the Timeline, what happens when you add a clip that overlaps another clip that's already in the track you're editing to depends on the Timeline > Layered Audio Editing setting. By default, with Layered Audio Editing turned off, overwriting one audio clip with another results in the overlapping part of the overwritten clip being non-destructively deleted from the Timeline by the incoming clip.

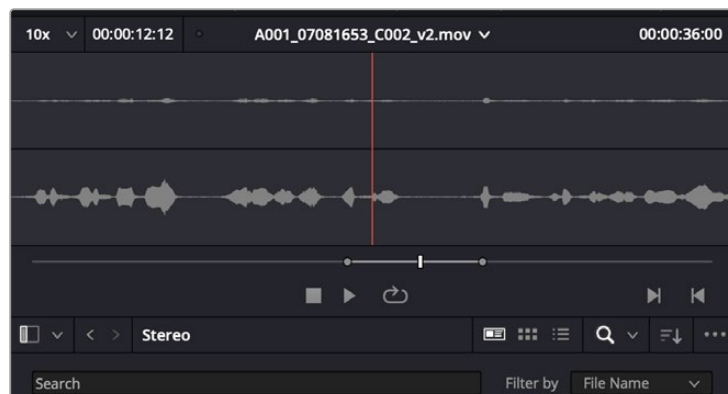
However, if you turn Layered Audio Editing on, then incoming clips do not overwrite overlapping clips in the Timeline; instead, they're layered within that track, such that the incoming audio clip takes precedence over what was previously there, but overlapping audio segments that were previously in the Timeline are preserved, which can be seen when you choose View > Show Audio Track Layers.

In this way, you can choose whether you want to overwrite previously edited clips, or layer newly edited clips, as your needs require, regardless of whether or not Audio Track Layers are visible. Audio layering can be enabled in both the Edit and Fairlight pages.

For more information about audio layering, see the section later in this chapter.

Choosing Parts of Clips to Edit in the Media Pool

The Media Pool has a preview player at the top that provides a place to open selected source clips in the Media Pool, play them, add marks to log them, and set In and Out points in preparation for editing them into the Timeline via drag and drop. The Media Pool Preview Player effectively acts as a Source monitor for editing in the Fairlight page.



The preview player in the Media Pool

- Various viewing controls populate the title bar at the top. A dropdown menu at the upper left lets you choose a zoom level for the audio waveform that's displayed. To the right of that, a Timecode window shows you the duration of the clip or the duration that's marked with In and Out points. Next to the right, a real-time performance indicator shows you playback performance. In the center the title of the currently selected clip is shown, with a dropdown menu to the right that shows you the most recent 10 clips you've browsed. To the far left, a Timecode field shows you the current position of the playhead (right-clicking this opens a contextual menu with options to change the timecode that's displayed, and to copy and paste timecode).
- The center of the Media Pool Preview Player shows you the waveforms in all channels of the currently selected clip, at whatever zoom level is currently selected.
- Transport controls at the bottom consist of a jog bar for scrubbing, Stop, Play, and Loop buttons, and In and Out buttons.

Dragging Audio Clips Into the Timeline

You can open the Media Pool in the Fairlight page, and edit audio clips into the Timeline in their entirety by dragging and dropping individual clips onto whichever audio track you want them to appear. You can drag any clip onto any track, regardless of whether or not the channels of the clip match the channel mapping of the track. However, if you edit a clip with more channels than a track has (for example, editing a stereo clip onto a mono track), only the channels supported by that track will be output, with all other channels in that clip being muted. If this happens, you can always remap the audio track by right-clicking the track header and choosing a new mapping from the Change Track Type To submenu.

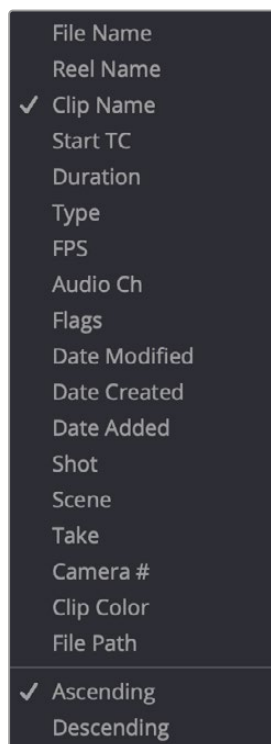
TIP: Dragging one or more clips to the empty area underneath the existing audio tracks of the Timeline results in the creation of new tracks, each of which is automatically mapped to however many channels are required by each audio clip being edited.

When working with Ambisonics files, dragging a clip to the timeline or creating a new timeline that refers to a clip creates a track of the appropriate type, matching the file's Ambisonic order and channels.

If you want to edit several clips into the Timeline at once by dragging them from the Media Pool, you'll probably want to do a bit of preparation to make sure they're edited in the right order.

Dragging several clips into the Timeline as one contiguous series of edited clips:

- 1 Change the sort order of the Media Pool's browser area to put the clips into the order in which you want them to appear. In Thumbnail view you can use the Sort Order menu, but in List view you can click the header of any metadata column to sort by that column's data.



Using the Sort Order menu to change the sort order of clips in the Media Pool

- 2 Use the Media Pool thumbnails, the Media Pool List view Filmstrip, or the Source Viewer to set In and Out points to define the part of each clip that you want to edit into the Timeline.

- 3 Select the Media Pool clips you want to edit into the Timeline by dragging a bounding box, Command-dragging multiple bounding boxes over different sets of clips, by Shift-clicking a range of clips, or by Command-clicking individual non-contiguous clips.

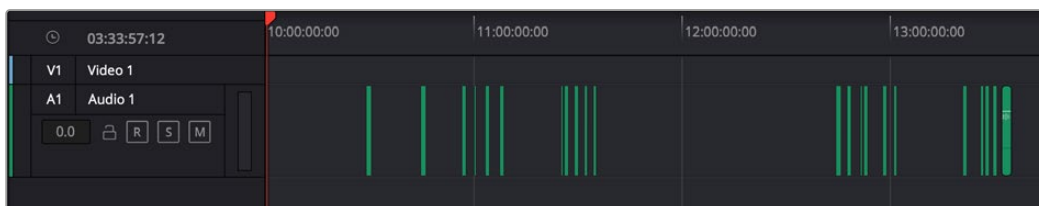
- 4 Drag any of the selected clips to the desired position in the Timeline to perform an overwrite edit.

The clip(s) you drag overwrite whatever other clips they overlap in the Timeline. Multiple clips dragged from the Media Pool will be edited in the order in which they're sorted in the Media Pool, using each clip's In and Out points.

It's possible to edit audio clips into the Timeline so that each clip's timecode lines up with the Timeline Ruler. This can be useful if you're organizing multiple source audio recordings that you want to synchronize together on multiple tracks.

Dragging multiple clips to edit them into a track at their timecode positions:

- 1 Select the Media Pool clips you want to edit into the Timeline by dragging a bounding box, Command-dragging multiple bounding boxes over different sets of clips, by Shift-clicking a range of clips, or by Command-clicking individual non-contiguous clips.
- 2 Hold Command-Shift down, and drag the selected clips into the track you want them to appear, to perform an overwrite edit.



A series of audio clips edited into the Timeline by timecode position

Each clip edited into that track appears at the same timecode position as its embedded timecode. This means that if you were recording time-of-day timecode, each clip will appear on the Timeline at the time it was recorded. A series of clips recorded during hour 10 through 13 will appear distributed throughout hours 10-13 on your timeline.

You can also edit two or more audio clips into the Timeline as a stack, in preparation for layering multiple sound effects for doing sound design work.

Dragging multiple clips to edit them into a track as a parallel stack:

- 1 Select the Media Pool clips you want to edit into the Timeline by dragging a bounding box, Command-dragging multiple bounding boxes over different sets of clips, by Shift-clicking a range of clips, or by Command-clicking individual non-contiguous clips.
- 2 Command-drag the selected clips into a track of the Timeline. The first of the selected clips appears in the track you're dragging to, the other clips appear either in audio tracks underneath the first one, or if there are no audio tracks available, in new audio tracks that will be created to hold those clips.

All clips you've edited appear as a parallel stack, in separate tracks, one on top of another.

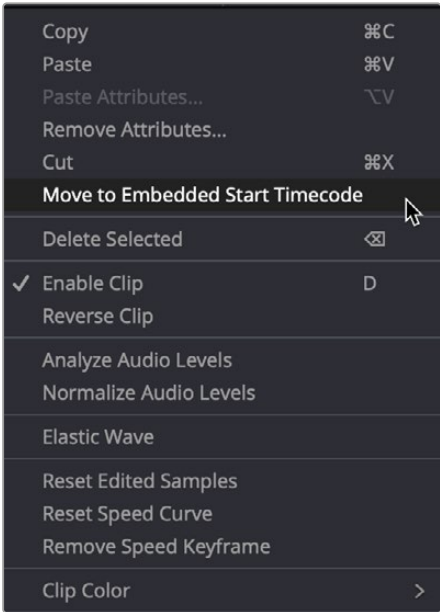
Moving Audio Clips to Embedded Timecode Position

All clips have a timecode embedded into their metadata. There are options to place clips onto the Timeline from either the Media Pool or in the Timeline itself with this metadata.

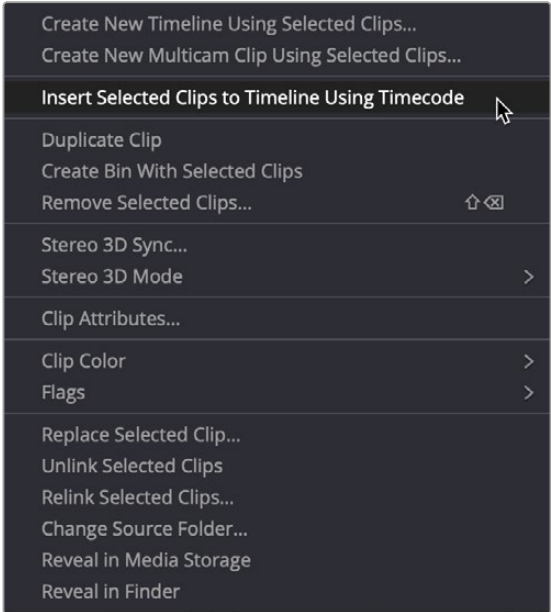
Right-clicking a clip in the Timeline has the choice to Move to Embedded Start Timecode. When this is clicked the clip will spot onto the Timeline on a selected track with the timecode.

It's important to be aware of the embedded audio timecode and the Timeline timecode. The two must have overlapping timecode, or this function will not work.

For example, a clip that has an embedded timecode of 00:00:00:00 may be difficult to find on a Timeline that starts at 00:59:58:00. If you have spotted a clip using these tools and still don't see the clip in question, then check the embedded timecode in the Metadata panel of the Inspector as well as the timecode of the Timeline itself.



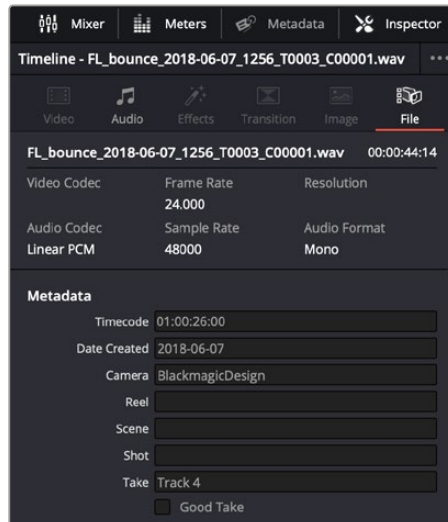
Right-clicking a clip in a Timeline reveals the Move to Embedded Start Timecode option.



Right-clicking a clip in the Media Pool reveals the Insert Selected Clips to Timeline option.

Right-clicking a clip in the Media Pool has the choice to Insert Selected Clips to Timeline Using Timecode. When this is clicked the clip will spot onto the Timeline on a selected track with the embedded timecode.

NOTE: You can find the embedded timecode for a clip in the Inspector in the File tab under timecode.

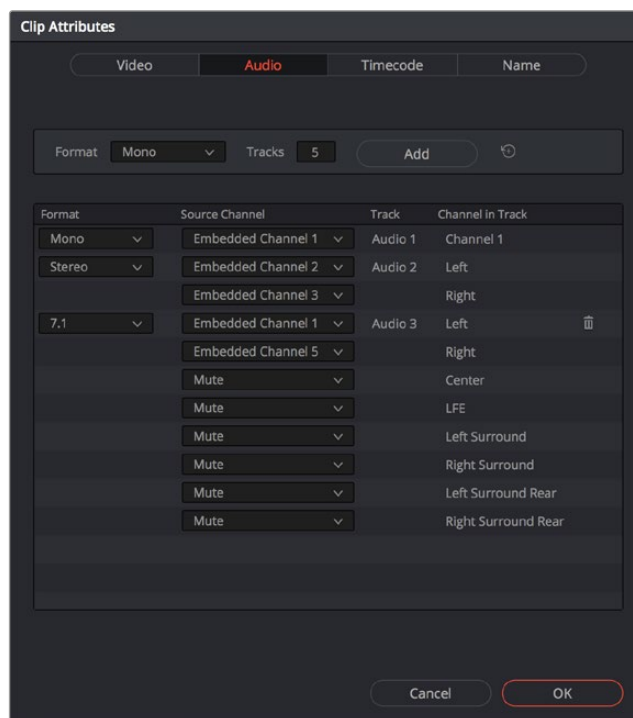


Timecode metadata is in the Inspector window under the File tab.

Support for Mixed Audio Track Formats from Source Clips

DaVinci Resolve supports media with multiple audio tracks that have differently formatted channels embedded within them. For example, a clip with one stereo track, one 5.1 surround track, and six mono tracks can all be appropriately set up in the Audio panel of Clip Attributes after that clip has been imported.

The Audio panel of Clip Attributes has controls over what format (Mono, Stereo, 5.1, 7.1, Adaptive) the channels embedded within a particular audio track should be configured as. This means you can set up clips with multiple tracks, each one using different formats of audio employing different combinations of channels, which is useful for setting up imported audio mix files that you want to output when mastering a program.



Clip Attributes now lets you assign channels among different tracks with different channel assignments

Making Audio Clip Selections in the Timeline

Nearly every editing operation described in this chapter and others requires you to make a selection to define which clips will be affected. Three editing modes in the toolbar give you different ways of selecting clips, depending on what you're trying to do, and how you like to work. These are (from left to right) the Pointer mode, the Range mode, and the Focus mode. Which mode you choose determines how clips and clip segments are selected in the Timeline in preparation for all manner of editorial operations.



The Pointer, Range, and Focus modes seen in the toolbar

Fairlight Edit Mode Remains Between Application Restarts

Fairlight retains the Edit Mode the project was saved with between restarts. Whatever edit mode you were working in prior to closing the application will be active when reopened.

Why Are There Three Edit Modes?

While the Pointer and Range modes can also be used with the pointer, they're really designed to enable automatic selections based on the position of the playhead. This is accomplished when using the Fairlight Editing console, the Fairlight Desktop Console, or keyboard shortcuts to control Timeline transport, while specific tracks are selected to enable selection and editing on those tracks.

The Focus mode is designed for efficient selections made using the pointer via a mouse, trackpad, or pen and tablet, made in conjunction with a variety of commands for extending and editing selections triggered via keyboard shortcuts. If you're editing with a keyboard and mouse, this mode is designed to let you work quickly by enabling a variety of different selection functions based on clicking different parts of clips.

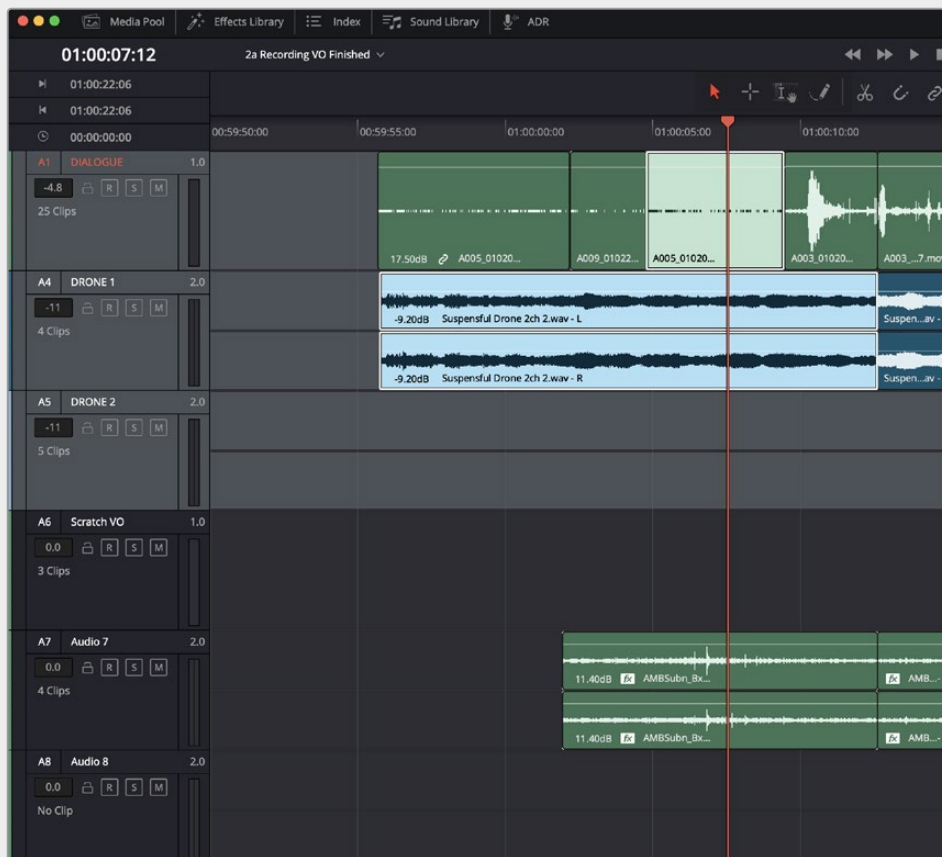
Selecting Tracks

In order to understand clip selection, you must first understand track selection. The Timeline on the Fairlight page lets you select entire tracks to facilitate the automatic selection of clips that intersect the playhead on those tracks using keyboard shortcuts, the Fairlight Desktop Console, or the Fairlight Editing panel, in Pointer and Range modes (described in upcoming sections).

For example, were you to select tracks A2, A3, and A4, then moving the playhead to intersect two clips on those tracks in Pointer mode automatically selects them, so they're ready for any operation you want to perform on both clips. To give a few examples, you could now split both clips at the playhead, Cut Head or Tail to the playhead, Delete both clips, or Copy them in preparation for pasting elsewhere.

Additionally, there are times when clicking or dragging on one or more clips with the pointer results in both those clips being selected, along with the tracks on which they sit. For example, selecting clips using the Focus mode will also select the tracks those clips are on.

If you're manually selecting tracks using the pointer, there are different ways to do so.



Command-clicking multiple track headers selects those tracks

Methods of selecting and deselecting tracks in the Fairlight page Timeline:

- **To select a single track:** Click anywhere in the background or on the track number of that track's header (not on a button). In Range mode you can also click in any unused area of the track itself.
- **To deselect a single track:** Click anywhere in the background or on the track number of a previously selected track's header (not on a button). In Range mode you can also click in any unused area of the track itself. If multiple tracks are selected, Command-clicking one will remove just that track from the selection.
- **To select multiple tracks:** Command-click in the track header background of every track you want to select. In Range mode you can also Command-click in any unused area of the tracks themselves. Command-clicking an already selected track will deselect it.
- **To select multiple continuous tracks:** Click anywhere in the background or on the track number of a track header, and then drag a bounding box up or down over all other tracks you want to select. In Range mode you can also drag a bounding box over any part of the tracks themselves, while also defining a range in which you want to work.
- **To move the selection to higher or lower tracks:** Press Control-Option-Up Arrow or Down Arrow to move the selection state to the next track higher (Control-Option-Up Arrow) or lower (Control-Option-Down Arrow). If multiple tracks are selected, then the multi-selection will be moved as a block; for example, selecting tracks A2 and A3 and then pressing Control-Option-Down Arrow will result in tracks A3 and A4 being selected.

Using Pointer Mode

Pointer mode uses the position of the playhead to make automatic selections when using the Fairlight Editing console or keyboard shortcuts to make clip selections on selected tracks. However, you can also use this mode in conjunction with the pointer and keyboard shortcuts to make selections in a different style. Pointer mode is primarily intended to allow efficient editing of whole clips.

- **If no tracks have been selected:** Clips that intersect the playhead are not selected. You can use the pointer to select one or more clips by clicking, Command-clicking, or dragging a bounding box. Clips you select in this way are highlighted in orange. Making selections in this way is similar to making clip selections in the Timeline of the Edit page.
- **If tracks have been selected:** All clips that intersect the playhead on selected tracks will be automatically highlight selected, but no In and Out points will be set. Clips on de-selected tracks will be ignored. Selecting one or more clips with the pointer (by Command-clicking or dragging a bounding box) creates an orange-highlighted selection.
- **If you set In and Out points:** Some functions will affect ranges of clips between In and Out points on selected tracks. Clips on unselected tracks are ignored.

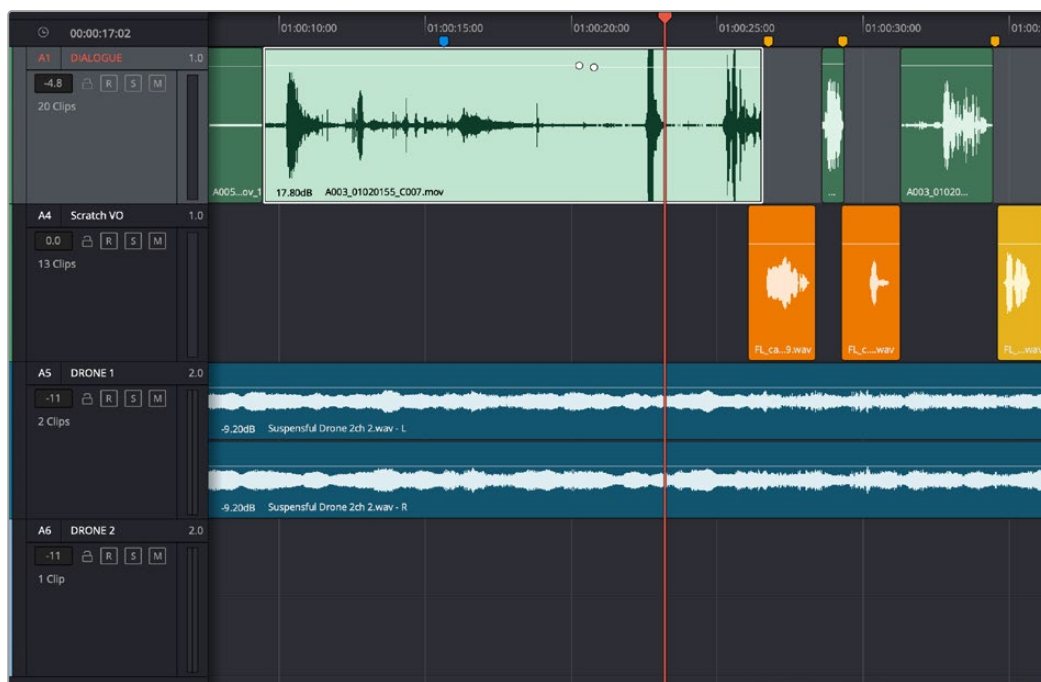
To choose Pointer mode:

- Click the Selection tool (the arrow) in the toolbar.
- Choose Trim > Pointer mode.
- Press A.

To automatically select clips using the playhead position in Pointer mode:

- 1 Press A to enter Pointer mode.
- 2 Select one or more tracks with clips you want to select.
- 3 Move the playhead to intersect those clips.

All clips that intersect the playhead on selected tracks of the Timeline are automatically selected in their entirety. Automatic selections are illuminated brighter to indicate their selection. Intersecting clips on unselected tracks are not selected.



Clips intersecting the playhead in Pointer mode are automatically selected with orange highlights

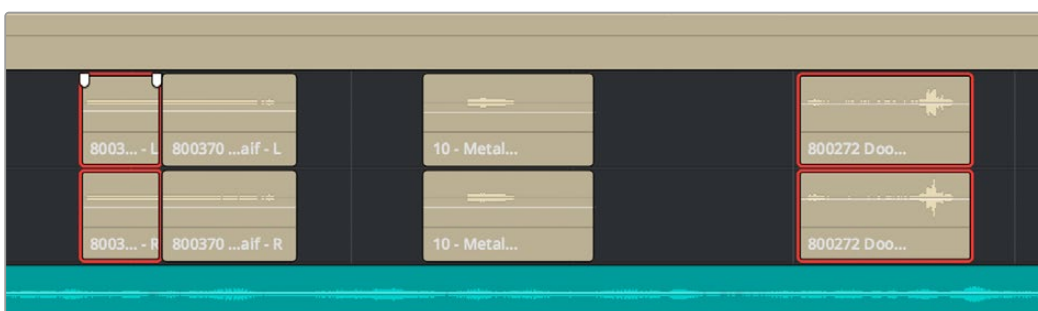
Methods of selecting clips using the pointer:

- Click any clip to select it.



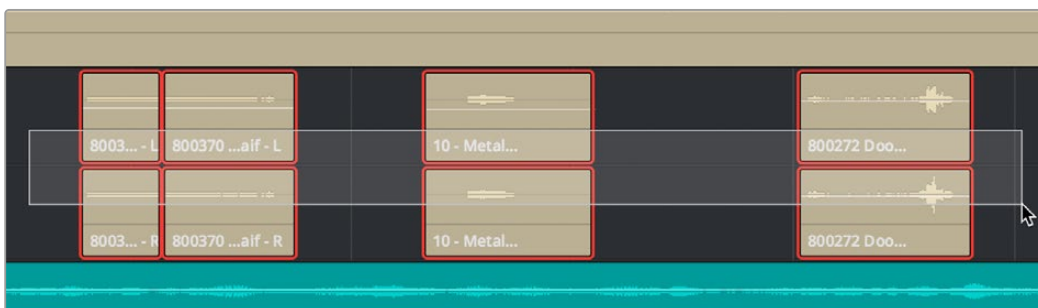
Click a clip to select it

- Command-click multiple clips to select them all at once.



Command-clicking multiple clips selects those clips, even if they're separated by other clips

- Click anywhere in the background of the Timeline and drag a bounding box over multiple clips.



Dragging a bounding box over multiple clips selects all of them

Using Range Mode

Range mode also uses the position of the playhead to make selections of a partial range of clips in the Timeline when using the Fairlight Editing console or keyboard shortcuts. You can also use this mode in conjunction with the pointer and keyboard shortcuts to make partial selections of clips.

- **If no tracks have been selected:** Clips that intersect the playhead are not selected. You can use the pointer to click a clip and select it in its entirety along with the track it's on. You can also use the pointer to drag on one or more clips to select a partial range in preparation for different editing operations. Whenever you make a selection with the pointer, Timeline In and Out points are set to the boundaries of the selection.
- **If tracks have been selected:** Selected tracks are brightened in the Timeline, and any clip on a selected track that intersects the playhead will be automatically highlighted brighter. Clips on de-selected tracks will be ignored. Dragging a crosshairs over one or more clips with the pointer

overrides all automatic selections and selects the regions of the clips you drag over, and the tracks they're on.

- **If you set In and Out points:** Partial regions of all clips on all selected tracks between the In and Out points will be highlighted brighter. Clips on unselected tracks are ignored. While In and Out points are set, the playhead no longer makes automatic selections; you must set new In and Out points to modify the selection in this mode.

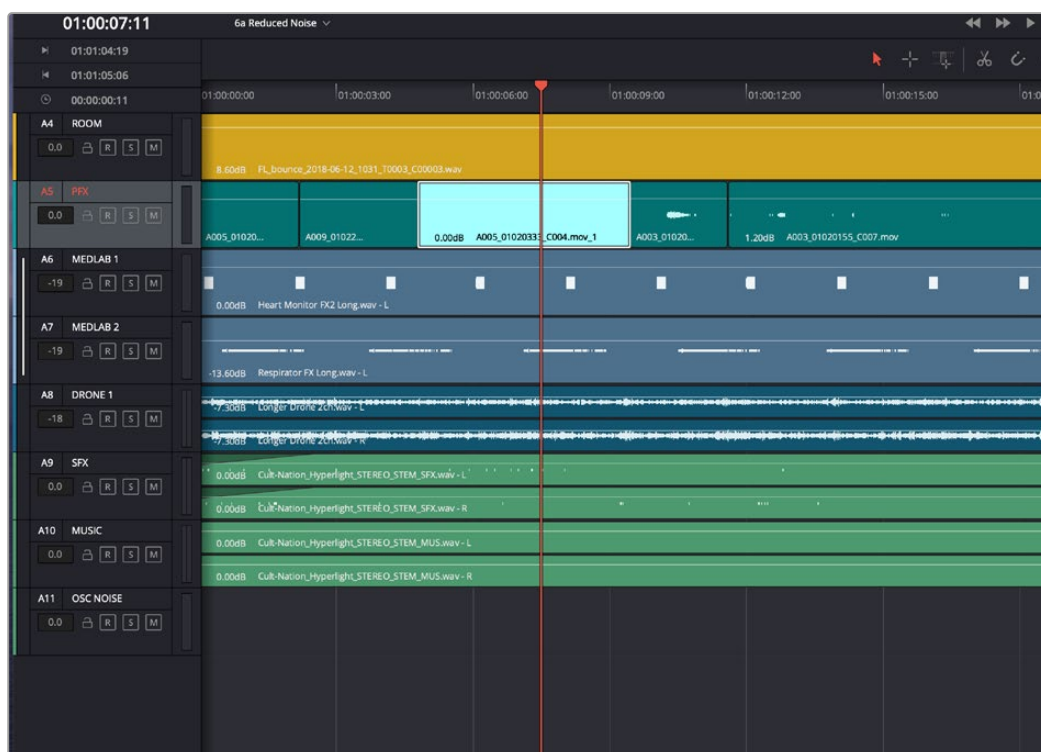
To choose Range mode:

- Click the Range Selection tool (the crosshairs) in the toolbar.
- Choose Trim > Range mode.
- Press R.

To automatically select clips in Range mode using the playhead position:

- 1 Press R to enter Range mode.
- 2 Select whichever tracks have clips you want to select.
- 3 Move the playhead to intersect those clips.

All clips that intersect the playhead, on the tracks you selected, define a selected range from the beginning of the first selected clip to the end of the last selected clip.

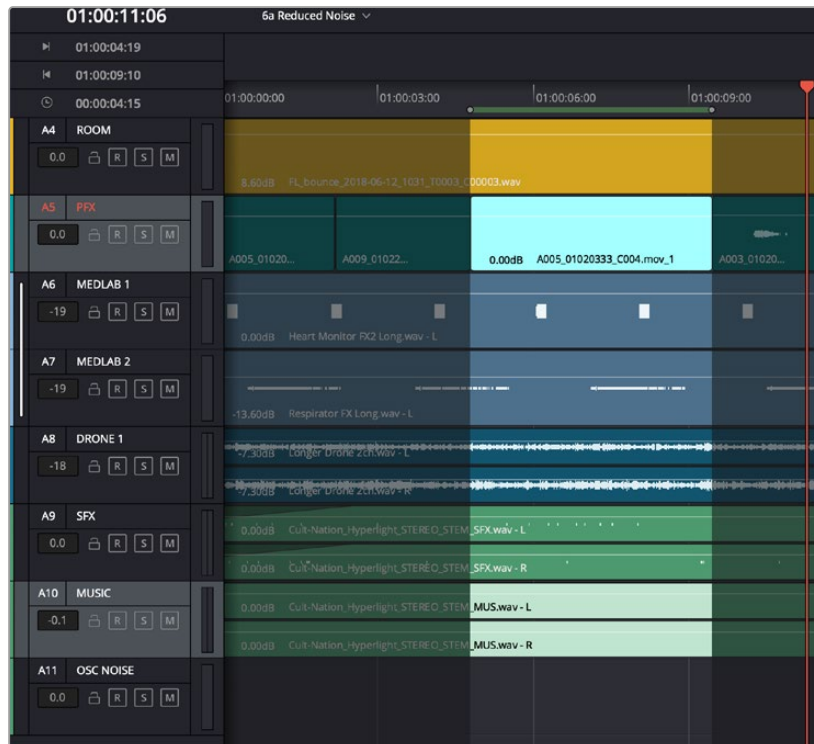


Clips intersecting the playhead on a selected track in Range mode are automatically selected

To create custom clip ranges using In and Out points in the Timeline:

- 1 Press R to enter Range mode.
- 2 Select whichever tracks have clips you want to select.
- 3 Move the playhead and press the I (Mark In) and O (Mark Out) keys to define a range in the Timeline.

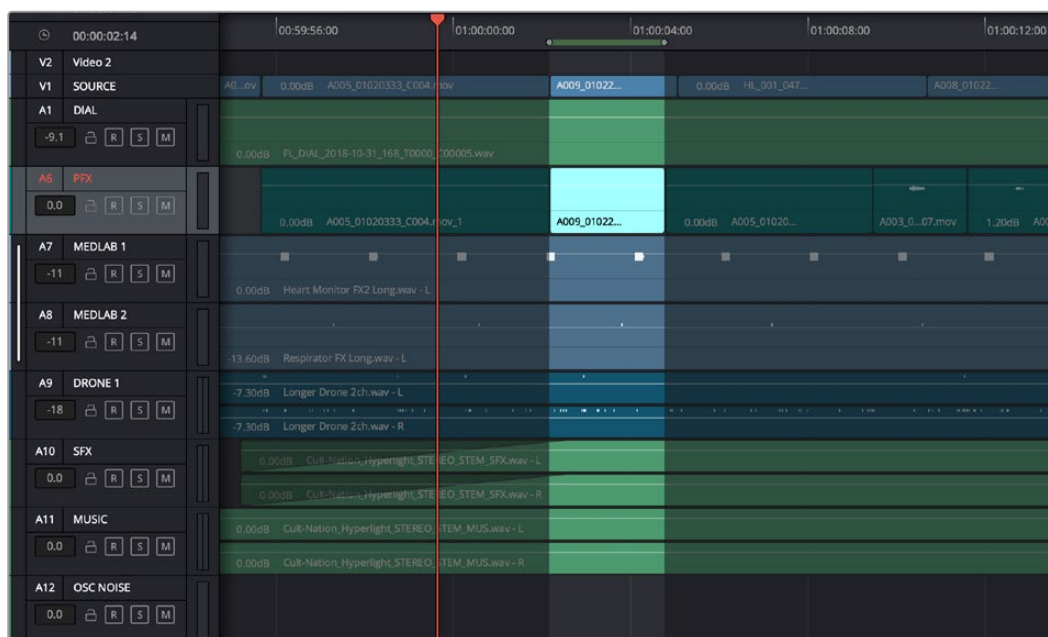
All clip segments that fall within the range of your In and Out points on selected tracks will be selected.



Using the Range mode to select clip segments on selected tracks using In and Out points

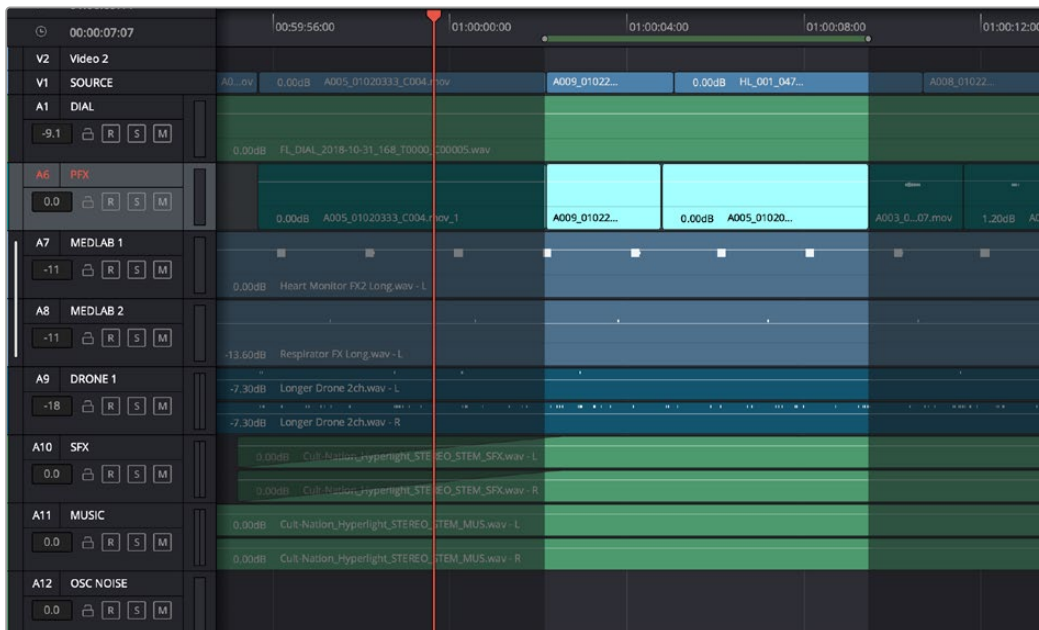
Methods of selecting clips in Range mode via clicking and dragging:

- **To select a single clip:** Click any clip to select both it and the track it's on, and define a range in the Timeline that matches the duration of that clip.



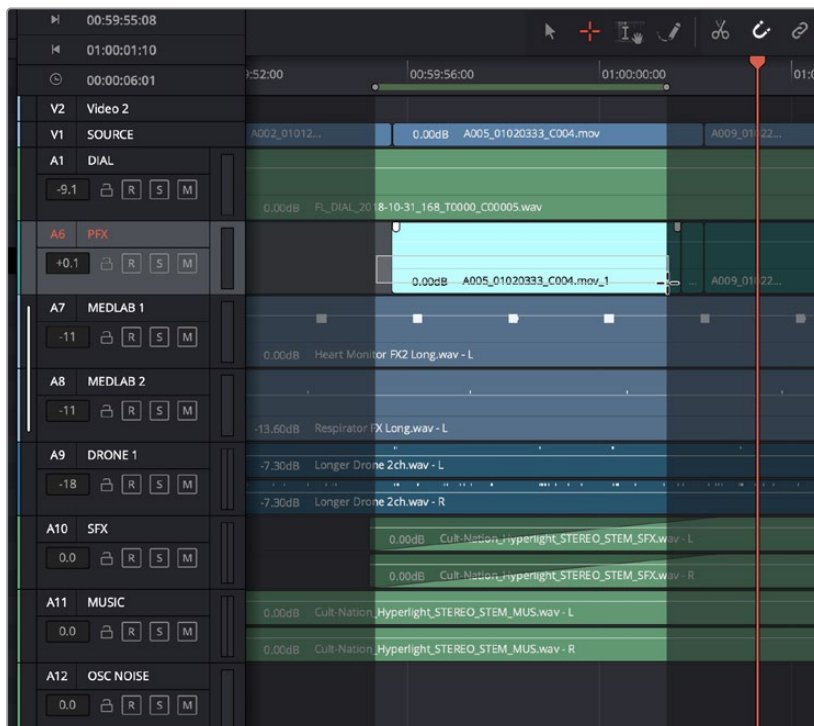
Click to select a single clip as well as the track it's on

- **To select multiple clips:** Command-clicking multiple clips to select them all at once defines a range in the Timeline that matches the total overlapping duration of all clips in the selection, from the beginning of the first selected clip to the end of the last selected clip.



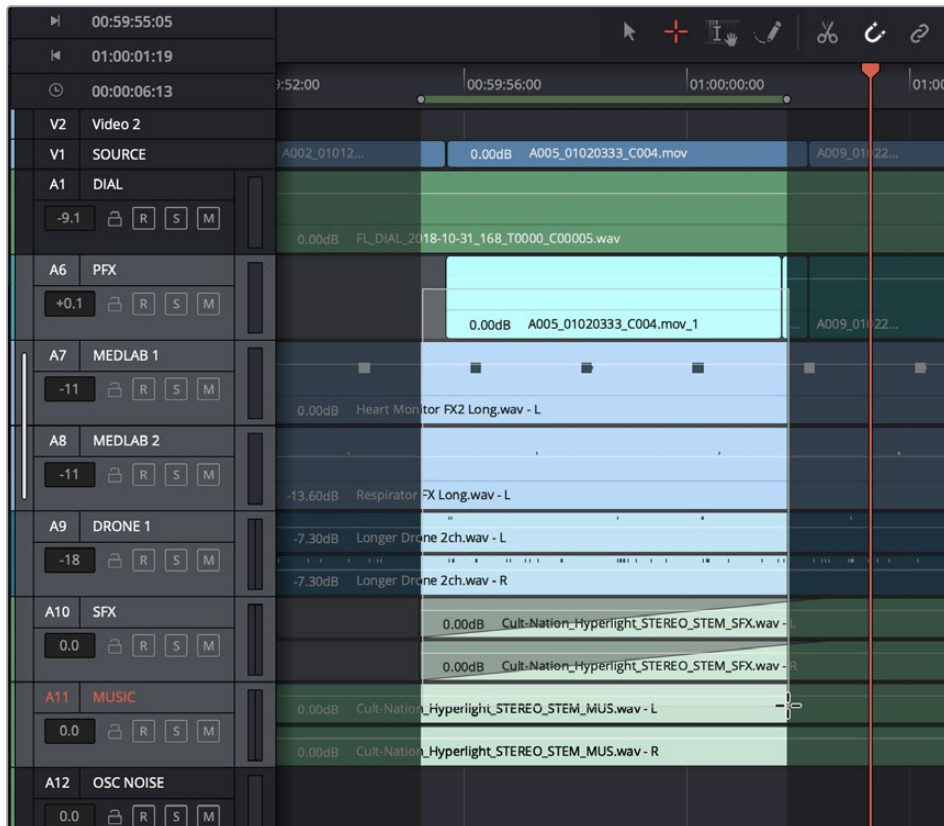
Range selection by Command-clicking multiple clips

- **To drag to select a range within a single clip:** With the Range Selection tool selected, drag anywhere on top of a clip to drag a bounding box over whatever segment of that clip (or of one or more clips) to select both that clip segment and the track it appears on. This is a good way of selecting part of a recording you want to move or delete.



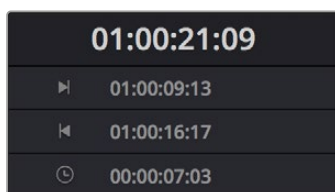
Ranged selection within a clip using a bounding box

- **To drag to select a range within multiple clips:** Click and drag a bounding box over whatever segment of one or more clips you want to select to select both those clip segments and the tracks they appear on. Or, Command-click and drag anywhere on top of any clip to drag a bounding box over whatever region of clips and tracks you like.



Ranged selection across multiple clips using a bounding box

In Range mode, whenever you make a selection, the In and Out point fields update with the range that you've created.



The range fields showing you the In and Out point values that define the current range

These ranges can be cleared if necessary.

Methods of clearing In and Out points to clear the current range:

- Press Option-I to clear the current In point.
- Press Option-O to clear the current Out point.
- Press Option-X to clear both the In and Out points.

Using Focus Mode

Focus mode works as a Multi Tool that is designed for making efficient pointer-based selections and edits, in conjunction with an assortment of commands for extending and editing selections that can be triggered via customizable keyboard shortcuts. If you're editing with a mouse and keyboard, this mode is designed to let you work quickly by enabling a variety of different functions based on clicking different parts of each clip in the Timeline.

Additionally, an important aspect of working in Focus mode is that this is the only mode that lets you edit the Timeline during playback, which you can't do in Pointer and Range modes.

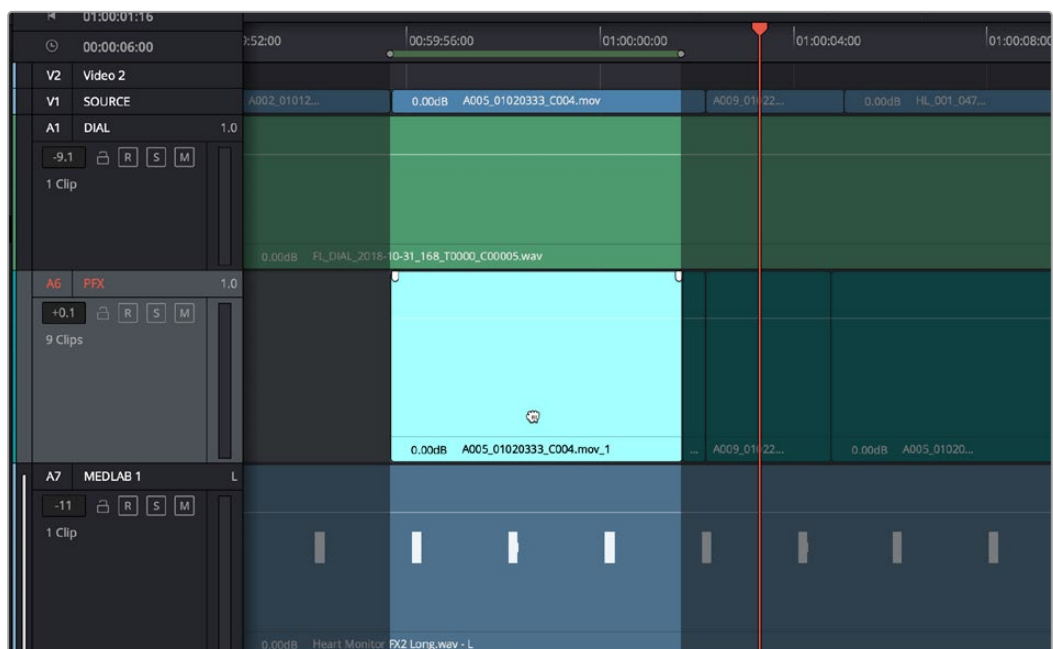
- **If no tracks have been selected:** Whenever you make a selection with the hand, Timeline In and Out points are set to the boundaries of the selection. Whenever you make a selection with the crosshairs, Timeline In and Out points are set to the boundaries of the region you dragged. In all cases, the tracks that contain selected clips and regions of clips are also selected.
- **If tracks have been selected:** Any clip on a selected track that intersects the playhead will be automatically highlighted brightly. Clips on de-selected tracks will be ignored. Dragging a crosshairs over one or more clips with the pointer overrides all automatic selections and selects the regions of the clips you drag over, and the tracks they're on.
- **If In and Out points have been set:** Clicking the bottom half of clips in the Timeline will select that clip and track, and the In and Out points will change to encompass that clip.

To choose Range mode:

- Click the Range Selection tool (the crosshairs at the bottom of a track) in the toolbar.
- Choose Trim > Range mode.
- The keyboard shortcut is pressing - R.

To select an entire clip using the Hand tool of the Focus mode:

- Move the pointer to the bottom half of a clip until a hand cursor appears, and click once to select that clip in its entirety.

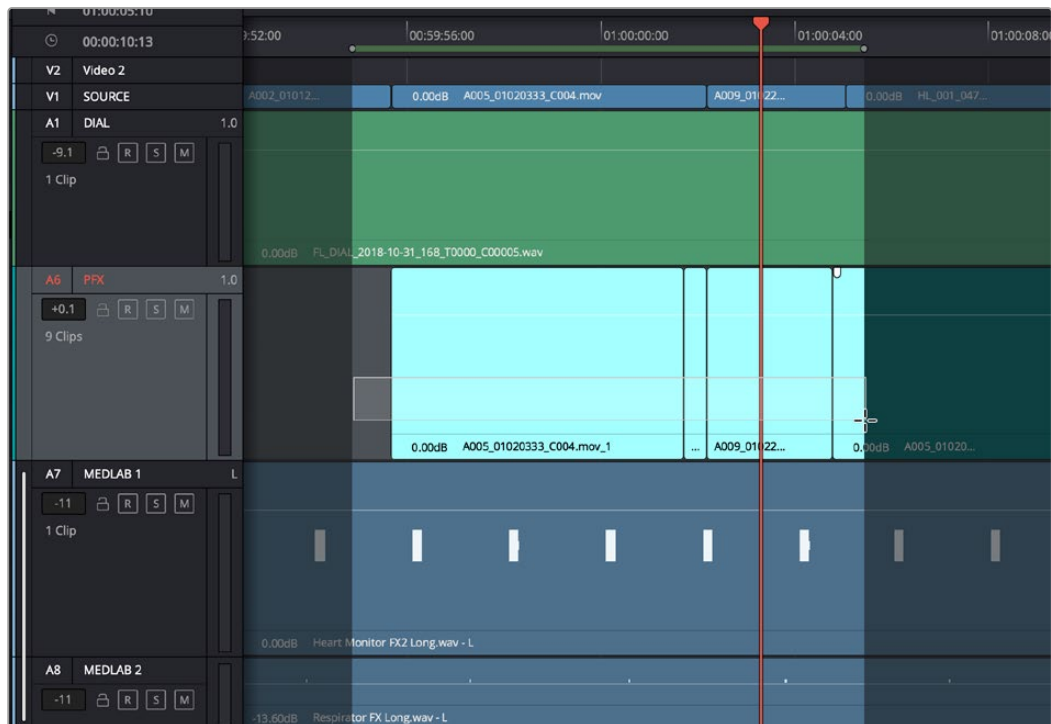


Clicking the bottom half of a clip to select that clip in Focus mode

- Use the Hand to Command-click multiple clips to make either contiguous or noncontiguous selections.
- Use the Hand to Shift-click multiple clips to make contiguous selections.

To select a range using the I-beam cursor of the Focus mode:

- Move the pointer to the top half of a clip until an I-beam cursor appears, and drag to select a region of one or more clips. As you drag, the playhead follows the Out point.

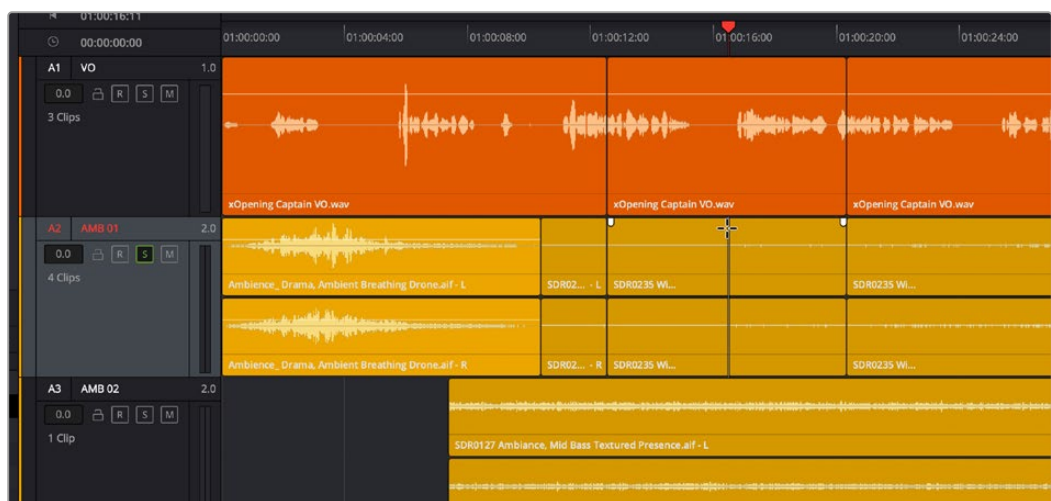


Dragging within the top half of a clip to select a region in Focus mode

- Using the crosshairs, you can also Shift-click to expand or contract the selected region across one or more clips.
- Using the crosshairs, you can also double-click to select an entire clip.
- Partial edit selections that contain a fade will copy with the fade intact.

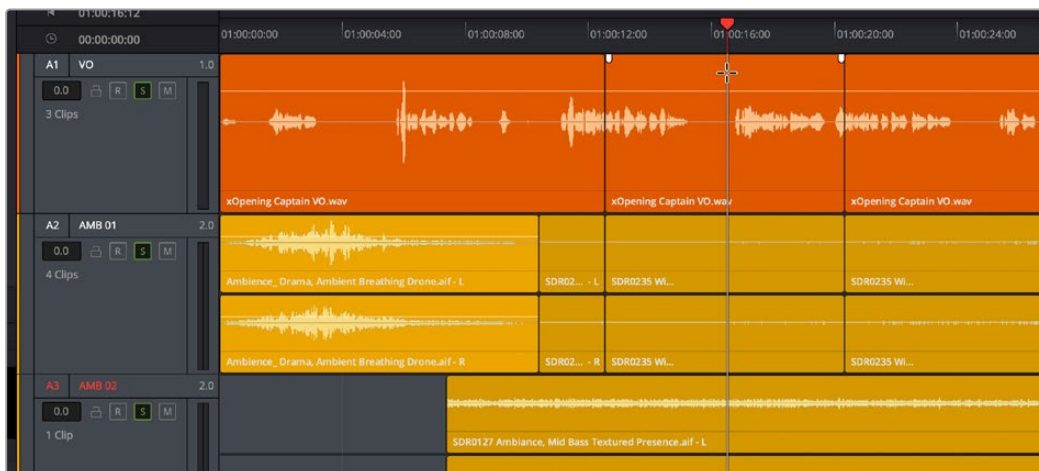
To select a single frame using the I-beam cursor of the Focus mode:

- Move the pointer to the top half of a clip until the I-beam cursor appears, and click once to place a point selection at the frame you clicked. The playhead also moves to this frame.



Clicking within the top half of a clip to make a single frame selection

- Using the I-beam cursor, you can also Command-click clips on higher and lower tracks to add them to the single frame selection, aligned at the same frame. For example, you could do this to split multiple clips at once. One frame selections can be on discontinuous tracks.



Command-clicking within the top half of multiple clips on multiple tracks to add single frame selections at the same frame

To move a selection using the Hand tool of the Focus mode:

Once you've made a selection of one or more clips or regions using the Hand or I-beam cursor of the Focus mode, you can move the pointer to the bottom half of any selected clip and drag the selection to another position on the Timeline.

Commands For Editing and Extending the Selection

Once you've made one or more selections in the Timeline, there are a series of commands you can use to modify or expand the selection. These commands were designed to be used alongside the Focus mode, but they can be used in any mode.

Editing the Selection

There are six commands for changing the current selection, which let you move it from one clip or group of clips in the Timeline to another. These commands only move the selected range of clips/frames; they do not move the clips themselves. These commands are located in the Timeline contextual menu under Edit Operations when you right-click on a clip.

- **Move To Previous/Next Edit:** Moves the current selection to the next clip/edit point to the left or right in the Timeline.
- **Move To Previous/Next Track:** Moves the current selection to the next track up or down in the Timeline.
- **Move to Previous/Next Frame:** Nudges the current selection to the left or right in the Timeline.

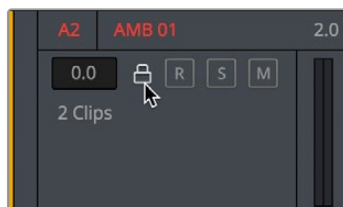
Extend Edit Selection

There are four commands for expanding the range of what's selected in the Timeline, one clip or track at a time. These commands are located in the Timeline contextual menu under Edit Operations when you right-click on a clip.

- **To Previous Edit:** Expands the selection to include the previous clip to the left in the Timeline.
- **To Next Edit:** Expands the selection to include the next clip to the right in the Timeline.
- **To Previous Track:** Expands the selection to include the clip in the next track up in the Timeline.
- **To Next Track:** Expands the selection to include the clip in the next track down in the Timeline.

Locking Audio Tracks

Another step you can take to prepare before performing any kind of editorial operation is to lock tracks with media that you don't want to be affected by whatever it is you're about to do. For example, if you have a complex set of music edits on track A3 that you don't want to be affected by operations that will delete media that overlaps it, you can lock track A3 so those clips remain unaffected.



Locking an audio track

Clips on locked tracks cannot be moved, deleted, cut, or otherwise affected by editorial operations. Furthermore, parameters of clips on locked tracks cannot be edited in the Inspector. However, clips on locked tracks can be played back and mixed like any other audio clips.

To toggle the lock or unlock state of audio tracks in the Fairlight page, do one of the following:

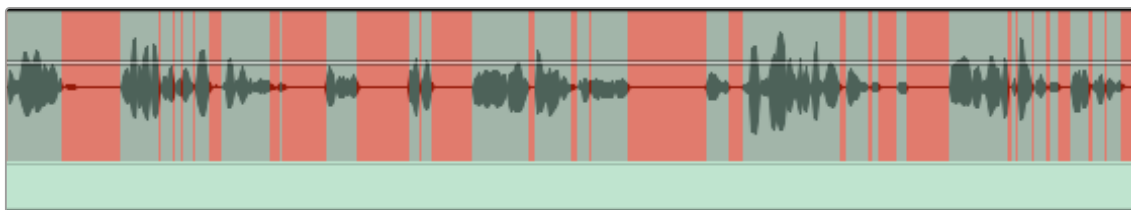
- Click any track's lock control to toggle lock on and off.
- Click any track's lock control and drag over the lock controls of other tracks in the Timeline to quickly lock or unlock several adjacent clips.
- Open the Index and click/drag one or more track lock controls to toggle lock on and off.

Splitting Clips

In addition to manually splitting timeline clips in either Pointer or Range mode, Fairlight offers two additional methods of managing your timeline audio. These are Remove Silence and AI Checkerboard to New Tracks, which, along with AI Create ADR Cues, comprise the Intellicut feature set. For more information about AI Create ADR Cues, see Chapter 172 “ADR (Automated Dialog Replacement).”

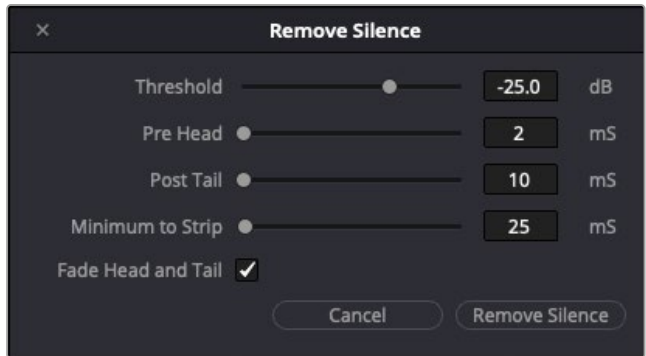
Remove Silence (Studio Version Only)

This feature identifies silent sections of a timeline clip and non-destructively removes them to increase the efficiency of your workflow.



Audio clip with silence indicators

Remove Silence Parameters



Remove Silence dialog

- **Threshold:** This determines the maximum signal level of your audio (in dB) that will be recognized as silence. Audio signals below the threshold value will be treated as silence and removed. Higher threshold values result in more audio being removed.
- **Pre Head:** Sets the amount of silence or pre-roll before each audible section.
- **Minimum to Strip:** This parameter determines the minimum amount of silence (in milliseconds) to be stripped from selected timeline clips.
- Higher values leave more silent sections in the audio. Lower values will result in more strips, but you may find that some of the remaining audio sounds “chopped up” because it abruptly cuts off.
- **Fade Head and Tail:** When this checkbox is filled, fades are applied to the beginning and end of each remaining clip based on the Pre Head and Post Tail values.

To remove silences from a timeline clip:

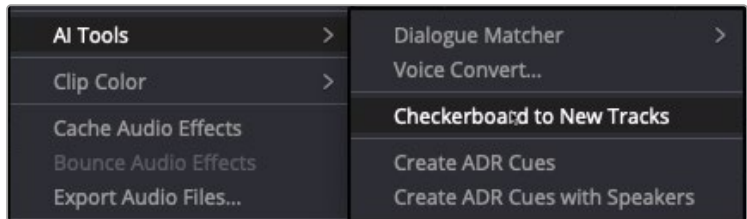
- 1 Select a clip or collection of clips on a track.
- 2 Right-click the selection and choose Remove Silence in the contextual menu.
- 3 The appearance of the selected audio will change to include vertical red stripes showing you the position and duration of the resulting silences based on the parameter values in the Remove Silence dialog.

AI Checkerboard to New Tracks (Studio Version Only)

If, for example, multiple actors were recorded with a boom mic on a single track and you need them on separate tracks, this feature will analyze and transcribe the original audio, then organize the resulting audio clips on separate tracks for each speaker’s voice.

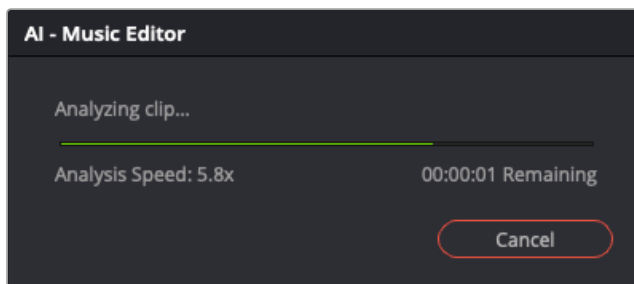
To split speakers to new tracks:

- 1 Right-click the timeline clip, or range of a clip, and choose AI Tools > Checkerboard to New Tracks in the contextual menu.



AI Tools contextual menu

- 2 When the Transcribing Audio dialog opens, Fairlight will use AI to analyze the audio and identify each voice.



Transcribing Audio dialog

- 3 A new track is created for each speaker, containing their new clips. The source timeline clip will be muted and left intact.



New track timeline clips for each speaker

NOTE: You may find it helpful to change the colors of the new timeline clips and tracks and timeline clips.

For more information on changing the color of a timeline clip, see *“Changing Clip Color in the Timeline,”* later in this chapter.

For more information on changing the track color, see *Chapter 170, “Using the Fairlight Page.”*

Manually Split Clips in Pointer or Range Mode

To split one or more clips in either Pointer or Range mode:

- 1 Select each track that has a clip you want to split.
- 2 Move the playhead to intersect the clips you want to split at the frame where you want the split to happen.
- 3 Do one of the following:
 - a) Choose Timeline > Split Clip or press Command-\ (backslash).
 - b) Choose Timeline > Razor or press Command-B.

When splitting a clip in the Edit page, a through edit appears to show that you currently have an edit with continuous timecode running from the outgoing to the incoming half. This is called a through edit and is displayed in the Edit page with a dotted line running along its edge so you know that it’s special. The Fairlight page doesn’t display through edits as of the time of this writing.

Linked Clips in the Fairlight Page

Clips can be linked together in the Fairlight page. When multiple clips are linked, Fairlight editing commands treat all linked clips as if they were a single clip. Anything you would do to a single clip is done to all linked clips at once. Selecting one item of a linked clip selects all items. Editing the In point of one item of a linked clip edits them all.

To link two or more clips together:

- 1 Select all clips you want to link together.
- 2 Right-click one of the selected clips, and choose Link Clips from the contextual menu.

A link indicator at the bottom left of every clip you've just linked shows their new linked status.

Trimming Clips Without Rippling the Timeline

Most basic adjustments in the Fairlight page only affect the selected clip or region of the Timeline. Clips to the right of the adjusted area of the Timeline are generally left alone so as not to inadvertently change sync when you don't expect it. This section covers the most basic parts of the Fairlight page's "seven-point editing" paradigm.

Multi-Point Editing Overview

Each clip in the Timeline has several draggable handles and click targets that let you perform different editing tasks using the pointer.

- **In point:** The left edge of the clip can be dragged to resize the beginning of the clip.
- **Out point:** The right edge of the clip can be dragged to resize the end of the clip.
- **Fade In handle:** A handle at the upper left-hand corner of the clip that only appears when the mouse is positioned over that clip, used to fade the audio in by dragging it to the right, or as part of a crossfade between two audio clips. Fades can be reset (eliminated) by double-clicking the fade handle.
- **Fade In curve:** A handle at the center of the Fade In curve that only appears when the curve is exposed, used to adjust the power of the fade in. This handle can be dragged vertically to change the X-Level of the fade, and horizontally to change the X-Point of the fade. Fade curves can be reset by double-clicking the curve handle.
- **Fade Out handle:** A handle at the upper right-hand corner of the clip that only appears when the mouse is positioned over that clip, used to fade the audio out by dragging it to the left, or as part of a crossfade between two audio clips. Fades can be reset (eliminated) by double-clicking the fade handle.
- **Fade Out curve:** A handle at the center of the Fade Out curve that only appears when the curve is exposed, used to adjust the power of the fade out. This handle can be dragged vertically to change the X-Level of the fade, and horizontally to change the X-Point of the fade. Fade curves can be reset by double-clicking the curve handle.
- **Level:** Considered an editorial characteristic, the level of any given audio clip can be adjusted via a level overlay running across each clip. The level of any clip can be reset to the default 0.0 dB by double-clicking the level overlay.
- **Position:** Clicking anywhere within the middle of a clip in Pointer or Range mode, or on the bottom of Focus mode, lets you drag that clip either forward or backward in time, or to another track.

Resizing the In and Out Points of a Clip

Trimming the head or tail of a clip in the Fairlight page means to resize the In or Out point of that clip, making it shorter or longer accordingly.

To resize the beginning or end of a clip:

- **To shorten or lengthen clips:** Move the pointer over the beginning or end of a clip, and when it turns into the Resize cursor, drag the In or Out point to the left or right to change the clip's length. As you drag the In or Out point of an audio clip in the Fairlight page, an overlay appears showing the waveform of all available media at the head (if you're dragging the In point) or tail (if you're dragging the Out point) of the clip you are resizing.



The overlay seen when resizing the In point of an audio clip in the Fairlight page

To quickly resize the beginning or end of a clip to the very beginning or end of available media:

- Double-click the In point of the clip to move the In point to the very beginning of that clip's media.
- Double-click the Out point of the clip to move the Out point to the very end of that clip's media.

If you resize a clip's In or Out point to overlap one or more neighboring clips in the Timeline, the overlapping parts of the neighboring clips will be overwritten by the clip you've resized.

Trim Start and Trim End

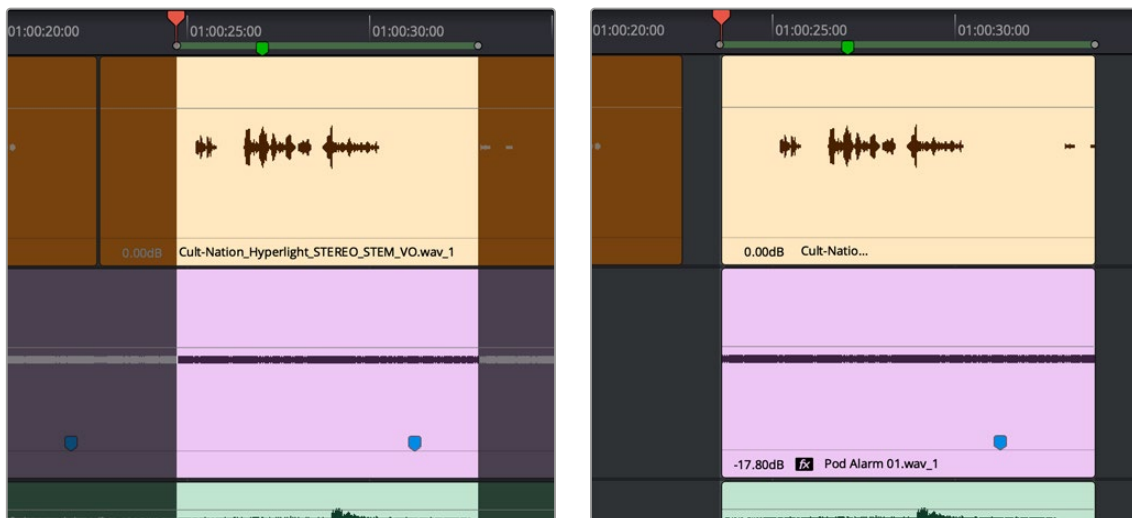
The Trim > Trim Start (Shift-I) and Trim End (Shift-J) commands let you move the In or Out point of all clips that intersect the playhead as either a ripple operation (in Trim mode) or a resize operation (in Selection mode). You do not need to make a selection to use Trim Start and Trim End, making these commands fast to use in the right situation. A classic use of Trim End is when you have several superimposed clips of different lengths that you want to either start or end at the same time.

- Trim Start resizes or ripples (depending on what mode you're in) all clips that intersect the playhead, so that each clip's In point is moved to the current playhead position.
- Trim End resizes or ripples intersecting clips so that each intersecting clip's Out point is moved to the current playhead position.

Clips that don't intersect the playhead are not affected. Furthermore, you can exclude clips on specific tracks from this operation by locking those tracks.

Trim to Selection

The Trim > Trim to Selection (Shift-Command-T) command simultaneously trims the heads and tails outside of a selection of one or more clips so that the selection is all that remains. This command is found within the Timeline contextual menu when you right-click on a clip.



(Left) A selection that includes parts of two clips, (Right) The result of using Trim to Selection to eliminate the heads and tails outside of this selection.

Moving and Overwriting Clips

There are a number of ways to move clips in the Timeline. Clips can be nudged and moved within a track, changing the timing of that clip's position in the edit or they can be moved up or down to other tracks.

To move clips in the Timeline, do one of the following:

- **To move one or more selected clips in the Timeline:** Drag any clip in the Timeline to any other position. If you're in Focus mode, you must drag using the bottom half of the selection. If you drag a clip to overlap another clip, the clip you're dragging overwrites the clip you're moving it over.
- **To nudge one or more selected clips in the Timeline by frame using the keyboard:** Make a selection, then press the Comma key (nudge 1 frame left) or Period key (nudge 1 frame right) to roll the selected edit to the left or right. Shift-Comma and Shift-Period nudges by 5 frames.
- **To move one or more selected clips up or down to other tracks at the same time:** Make a selection, then hold the Shift key down while dragging one of the selected clips up or down in the Timeline to lock their position in time while moving them to other tracks. Or, you can hold the Option key down and press Up or Down Arrow.

Sync Offset Indicator

Audio clips in the Fairlight page display an "out-of-sync" or sync offset indicators when they're moved out of sync with the video items they're linked to.

Snap to Playhead

While in Focus Mode, you can snap the head (beginning) or tail (end) of an audio clip to the playhead location (destination). This is useful when you want to line up audio with an onscreen event, such as a ball smashing a window.

Although “Automation follows edit” works with this function, there are some things to consider:

- This function works even if the playhead is offscreen.
- Clips can only be snapped individually and to a location on the same track.
- A “snapped” clip overwrites existing audio at the destination, even if the affected audio is on the top layer of a layered clip.
- A group of clips on a single track selected in Range mode using the playhead position can also be a destination. However, you would need to snap the clip to the side of the playhead that doesn’t have pre-existing audio you want to keep. For more information Range Mode, see “Using Range Mode.”

To Snap to the clip head to the playhead:

- 1 Activate Focus Mode and move the playhead to the intended destination.
- 2 Hold down Command-Option and click the clip you want to move.

The Audio is now in the new location with the head of the clip snapped to the playhead.

To Snap to the clip tail to the playhead:

- 1 With Focus Mode active, move the playhead to the intended destination.
- 2 Hold down Command-Option-Control and click the clip you want to move.

The Audio is in the new location with the tail of the clip snapped to the playhead.

Subframe Nudging

In the Preferences/User/Editing panel under General Settings you can change the nudge amount by either subframes or by milliseconds for the Fairlight page. These are completely user definable so that you can type your desired nudge amount into the corresponding box.

Be sure to fully investigate this settings panel, which includes pre-roll and post-roll settings and other useful Fairlight adjustments.

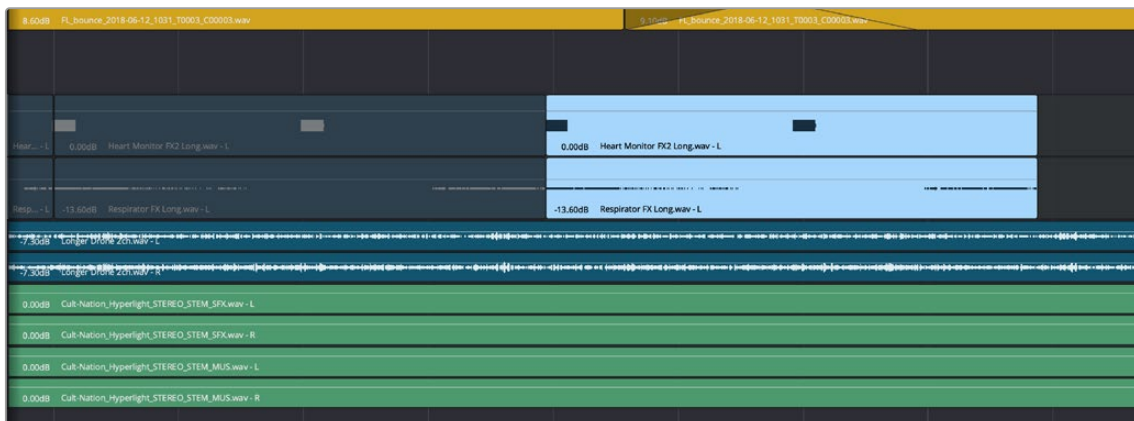
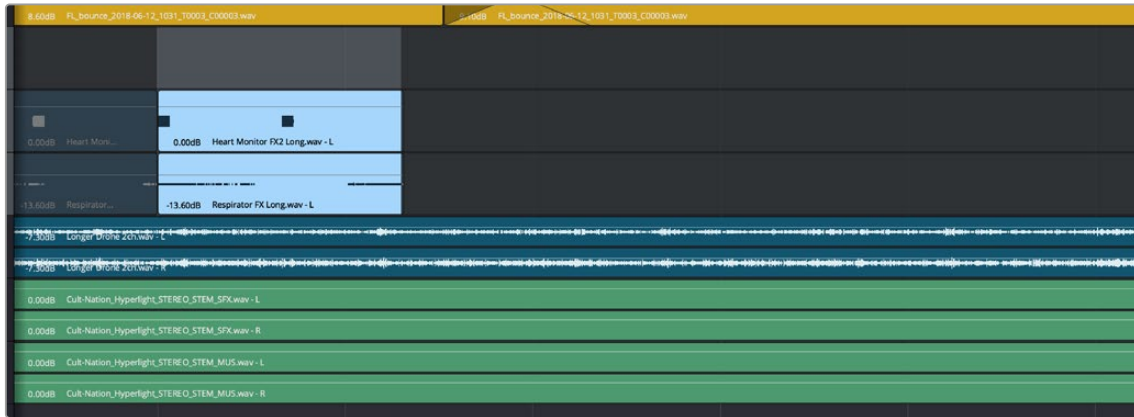
Slipping

Formerly called re-syncing, slipping an audio clip keeps that clip in the same place in the Timeline while changing the range of media that appears in that spot. Slip edits do not change the duration of the overall Timeline, and they don’t move the clip’s position relative to the other clips in the Timeline. Slipping simply changes the range of media that clip represents.

NOTE: While available in previous versions of Fairlight, slipping is not available at the time of this writing.

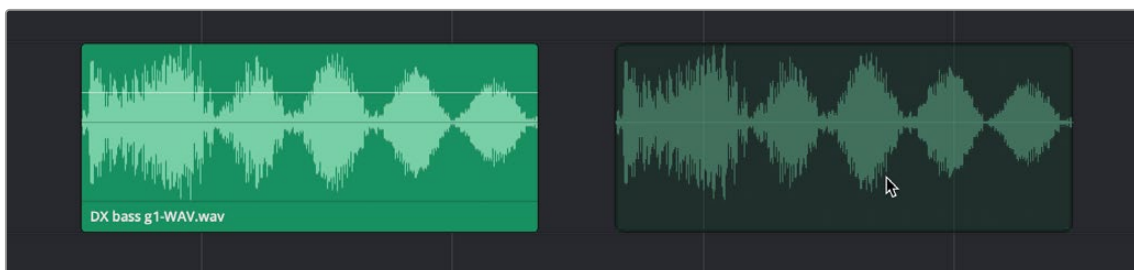
Duplicating Clips

The Duplicate Selection command duplicates one or more selected clips, placing the duplicates immediately after the Out point of the selection. If you're duplicating a region of a clip with tails, or a clip that's in between other clips, the duplicated selection will overwrite whatever is to the right of the current selection, for the duration of the duplicate.



Using the Duplicate Selection command to duplicate a selected region

If you hold the Option key down while dragging a clip in the Timeline, you'll place a duplicate clip wherever you drop it.



Option-dragging to duplicate a clip in the Timeline

Disabling and Re-Enabling Clips in the Timeline

Sometimes there's one or more audio clips in the Timeline that you don't want to play along with the rest of the edited sequence, but you don't want to remove from the Timeline either, in case you change your mind later. For this reason, it's possible to disable clips, effectively turning them off without removing them. Previous versions of Fairlight referred to this operation as Mute clip.

Disabled clips appear dimmed in the Timeline. They don't play back, they're not rendered, and they're not output to video. However, their position is preserved in the Timeline, so you can always re-enable them at a later time if you change your mind and decide you want to use them.

Disabled clips appear in the Timeline in gray.

To disable or re-enable one or more selected clips:

- Right-click part of the selection and choose Enable Clip from the contextual menu.
- Choose Clip > Enable Clip.
- Press D.

Deleting Audio Clips and Regions

You can delete any clips or clip regions that are selected, in either Pointer or Range modes, by pressing the Delete key, or by right-clicking a clip and choosing Delete Selected from the contextual menu. By default in the Fairlight page, deleting anything leaves a gap. Under the Edit Menu > Ripple Delete you can ripple delete selected clips changing the placement of all of the clips that follow.

Cut, Copy, and Paste

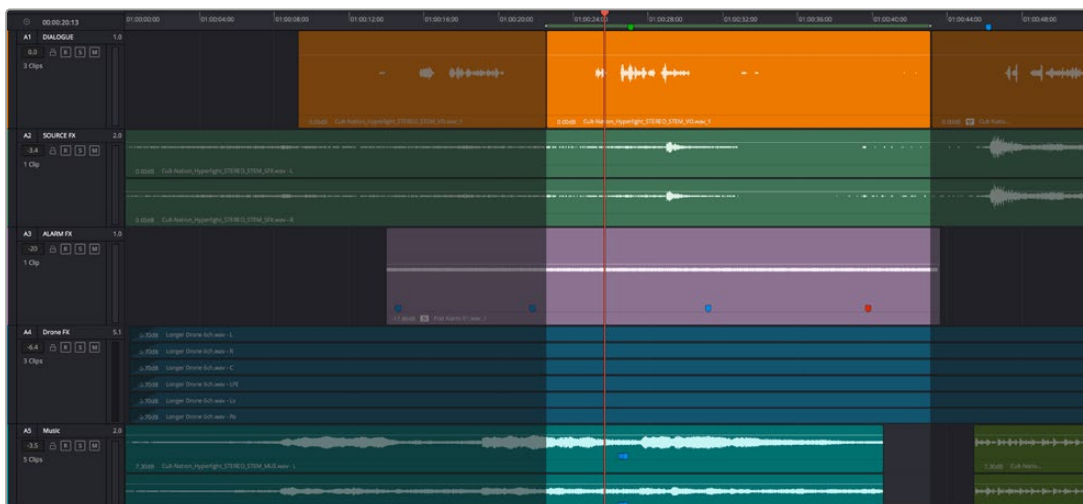
The Fairlight page has a unique copy and paste methodology that takes advantage of the "ghost" overlays that are used for the waveforms of selected audio clips. This method makes it easy to copy and paste clips using keyboard shortcuts and the JKL keys.

Conventional Cut, Copy, and Paste

The typical cut, copy, and paste commands expected of every software application are available in the Fairlight page, but with a unique twist that's particularly advantageous for users of the Fairlight control surface, or for anyone who uses the JKL transport key shortcuts to move around the Timeline for keyboard-driven editing.

To cut or copy and paste either all or part of a clip:

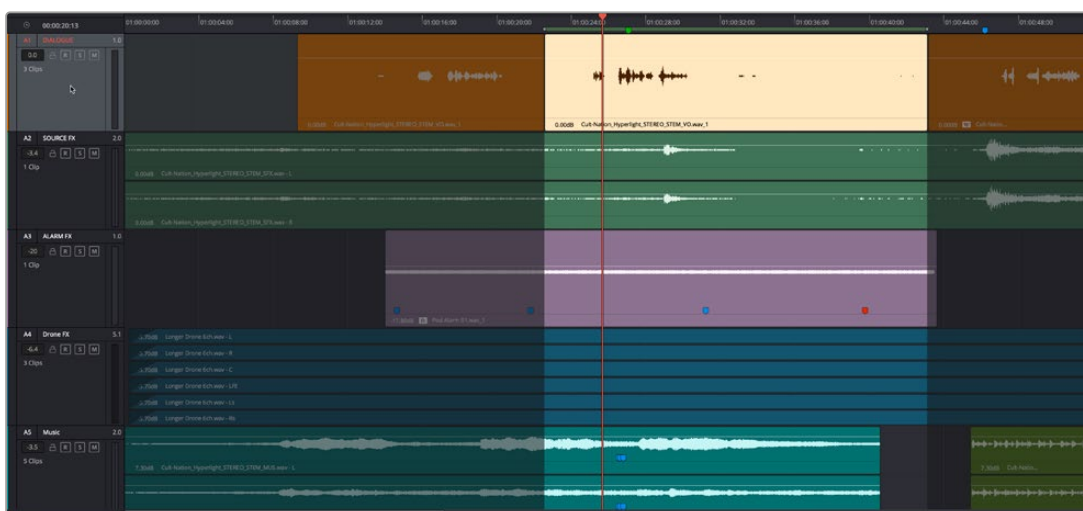
- 1 If you're cutting or copying a whole clip, then choose either the Selection (press A) or Range Selection (press R) mode. If you're cutting or copying part of a clip, then make sure you're in Range mode (press R).
- 2 To use the playhead to make a clip selection, select the track that contains the clip you want to copy or cut. If one or more tracks are already selected, you can use the Control-Option-Up or Down Arrow key shortcuts to move the track selection state up or down to the tracks with the clip you want to cut or copy.



Selecting the track with a clip you need to cut

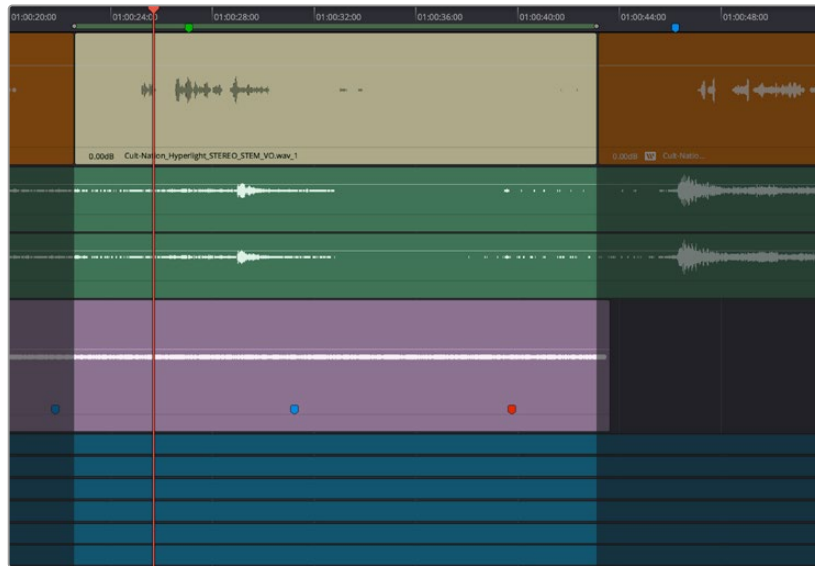
3 Do one of the following:

- a) To cut or copy a whole clip, move the playhead so that it intersects the clip you want to cut or copy. If the playhead intersects a clip on a selected track, that clip should become selected. You should note that even if you use the mouse to select a clip without selecting a track first, you should still move the playhead to intersect the clip you're copying or pasting, as this sets up an important reference point for the operation.
- b) To cut or copy a segment of a clip, move the playhead so that it intersects the clip you want to cut or copy. If the playhead intersects a clip on a selected track, that clip should become selected. Then, using JKL and the I (In) and O (Out) keys, mark a range in the Timeline that includes the segment of the clip you want. That segment should appear highlighted as a result.



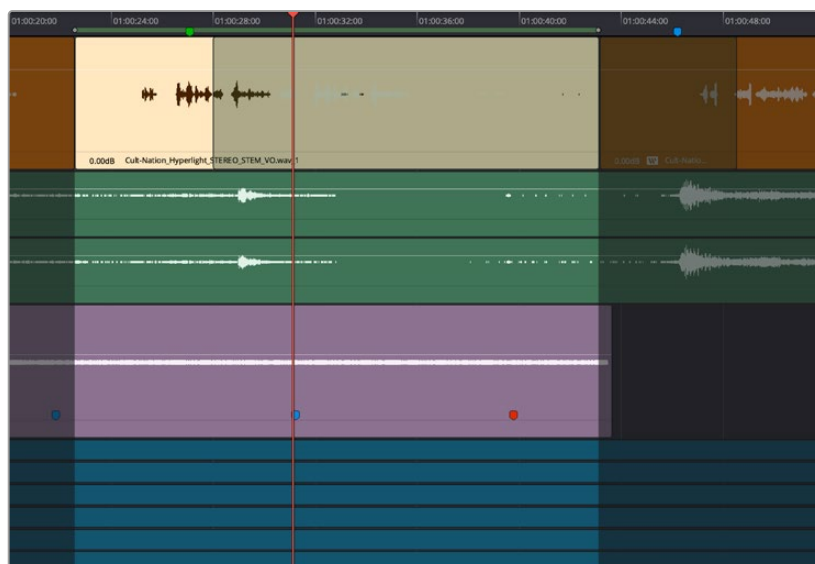
Cutting or copying a segment of a clip using In and Out points

- 4 Making sure the playhead is over the section of the waveform that you want to use as the frame to move the clip by, press Command-X to cut or Command-C to copy that clip (you can also right-click a clip and choose Copy or Cut). That clip will immediately become highlighted.



Cutting or copying a clip at the position of the playhead

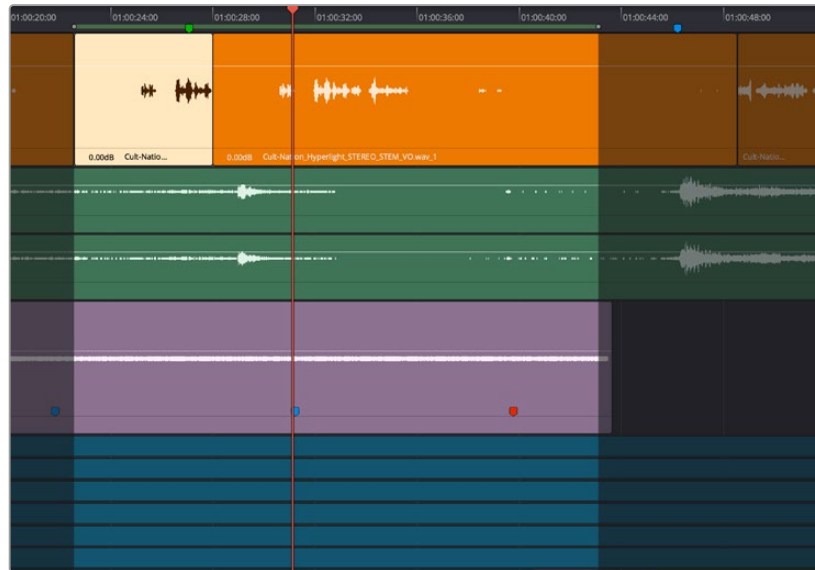
- 5 At this point, there are two things you can do to position the cut or copied clip to the position in the Timeline at which you want to paste it:
 - a) Moving the playhead will now also move the clip you cut or copied, shown as a ghost clip with waveform that's “attached” to the playhead at the frame you chose. Whether you drag the playhead with your mouse or use JKL to move the playhead through the Timeline, the cut or copied clip will move along with it, so that moving the playhead repositions the cut or copied clip on that track.
 - b) If you want to move the cut or copied clip to another track, use the Control-Option-Up or Down Arrow key shortcuts to change the selected track; the ghost clip will move along with change in track selection.



Positioning the cut or copied clip before you paste it

This way, you can use the playhead to align the ghost waveform with other audio clips surrounding it in preparation for pasting it.

- When the clip is positioned where you want it, press Command-V to paste the clip at the position you've chosen (you can also right-click a track and choose Paste from the contextual menu). The clip becomes solid, and you're finished.



The pasted clip

Using this method of cutting and pasting makes it quick to cut or copy clips using only keyboard commands, with the clip's ghost overlay making it easy to precisely align the clip you're pasting to fit exactly where you need it to, perfectly in sync.

Using the Cut/Copy Head and Tail Commands

Four additional commands make it easy to cut or copy portions of one or more clips that intersect the playhead, either from the In point to the current position of the playhead (the Head), or from the current position of the playhead to the Out point (the Tail).

To cut or copy the head or tail of a clip, and paste the result:

- Using these commands, there's no need to make a partial selection, so you can use either the Selection (press A) or Range Selection (press T) modes.
- Select the track that contains the clip you want to copy or cut. If one or more tracks are already selected, you can use the Control-Option-Up or Down Arrow key shortcuts to move the track selection state up or down to the tracks with the clip you want to cut or copy.
- Move the playhead so that it intersects the clip you want to cut or copy at the frame you want to define, either the end of the head or the beginning of the tail. If the playhead intersects a clip on a selected track, that clip should automatically become selected. You should note that even if you use the mouse to select a clip without selecting a track first, you should still move the playhead to intersect the clip you're copying or pasting, as this sets up an important reference point for the operation.
- Choose Edit > Cut/Copy Head/Tail to cut or copy the portion of the selected clip you want to paste. That portion of the clip will immediately become highlighted.
- At this point, there are two things you can do to position the cut or copied head or tail of the clip to the position in the Timeline at which you want to paste it:

- a) Moving the playhead will now also move the clip you cut or copied, shown as a ghost clip with waveform that's "attached" to the playhead at the frame you chose. Whether you drag the playhead with your mouse or use JKL to move the playhead through the Timeline, the cut or copied clip will move along with it so that moving the playhead repositions the cut or copied clip on that track.
- b) If you want to move the cut or copied clip to another track, use the Control-Option-Up or Down Arrow key shortcuts to change the selected track; the ghost clip will move along with change in track selection.

This way, you can use the playhead to align the ghost waveform with other audio clips surrounding it in preparation for pasting it.

- 6 When the clip is positioned where you want it, press Command-V to paste the clip at the position you've chosen (you can also right-click a track and choose Paste from the contextual menu). The clip becomes solid, and you're finished.

Paste and Remove Attributes

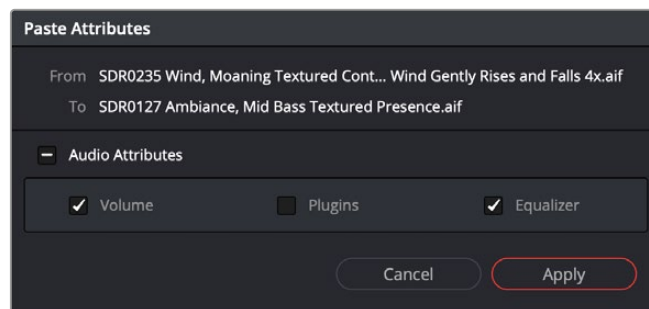
The Fairlight page has Paste Attributes and Remove Attributes commands that allow for the copying and resetting of audio parameters and effects, similar to the same commands on the Edit page.

Clip Attributes Naming

By double-clicking any clip in the Timeline, you can access the Clip Attributes window to rename the clip. You can still access this window with a right-click, revealing the dropdown menu of options.

Copying and Pasting Clip Attributes

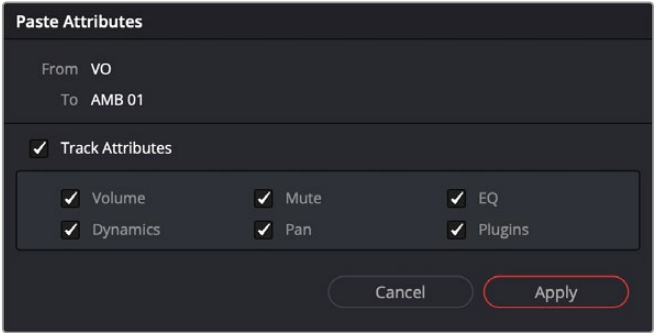
For clips, this works as simply as copying a clip, then selecting one or more audio clips and right-clicking another clip and choosing Paste Attributes from the contextual menu. A dialog appears letting you choose which audio attributes you want to paste before clicking Apply.



The Paste Attributes dialog in the Fairlight page

Copying and Pasting Track Attributes

For tracks, it works a little differently. Right-click on a track header and choose Copy Attributes to copy all track settings and effects. Then, select one or more other track headers, right-click the selection, and choose Paste Attributes. A dialog appears letting you choose which track attributes you want to paste before clicking Apply.



The Paste Attributes dialog in the Fairlight page, used to copy attributes from one track to another

Removing Attributes

For either clips or tracks, you can right-click and choose Remove Attributes to open a dialog with which to choose which attributes you want to reset to their default settings.

NOTE: By choosing the Volume check box in Remove Attributes you will delete all of the Clip Gain keyframes that have been added to a clip.

EQ, Level, and AI Dialogue Matching

To help provide a solid foundation for a great mix, Fairlight offers three features for matching the tonal characteristics (EQ profile), volume level, or sonic characteristics of disparate timeline clips as described below.

EQ Matcher

EQ Matcher can be used in situations where, for example, you need to quickly apply the tonal characteristic (EQ Profile) of on-location dialogue recorded with one type of microphone to newly recorded dialogue recorded in post using a completely different microphone.

To capture and apply an EQ Profile:

- 1 Right-click the timeline clip you want to match and choose Clip Operations > EQ Matching > Capture EQ Profile from the contextual menu.

Clip Operations	>	EQ Matcher	>	Capture EQ Profile
Elastic Wave	>	Level Matcher	>	Apply to Clip EQ
AI Tools	>	Analyze Audio Levels		Apply to Track EQ
		Normalize Audio Levels		Apply to Automated Track EQ

Timeline clip EQ Matcher contextual menu

- 2 Apply the captured EQ Profile to another audio clip in one of the following ways:
 - Apply to Clip EQ:** This applies the same EQ setting to the Clip EQ plugin for each selected clip on the destination track.
 - Apply to Track EQ:** Applies the EQ profile to the whole track. The applied EQ curve will appear in the EQ section of the corresponding Mixer channel strip.



New EQ curve displayed in a Mixer channel strip

Apply to Automated Track EQ: This option is useful when you have multiple clips on a track, each requiring its own unique EQ setting to maintain the correct tonal match during playback. This selection lets you quickly write automation data, which adjusts the Track EQ curve for each clip as required.

To apply the EQ profile to an automated track EQ:

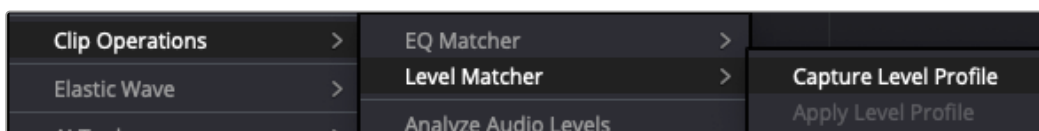
- 1 Right-click the timeline clip you want to match and choose Clip Operations > EQ Matching > Capture EQ Profile from the contextual menu.
- 2 Ensure the Toggle Automation button to the right of the transport control is active (red), then select an EQ parameter from the Automation drop-down in the destination Track Header, for example, EQ > Band 3 > Gain.
- 3 Right-click the first clip you want to apply the EQ profile to and choose Clip Operations > EQ Matching > Apply to Automated Track EQ. Repeat this step for subsequent clips on the track.
 - As you repeat the this step, you will notice the automation curve changing with each application of the EQ Profile.
 - During playback, the EQ Curve display in the corresponding mixer channel strip will change based on the automation data and Playhead position.

Level Matcher

Level Matcher lets you quickly and accurately match the volume of disparate timeline clips to each other.

To match your clip levels with the Level Matcher:

- 1 Right-click on the timeline clip you want to match and choose Clip Operations > Level Matching > Capture Level Profile.

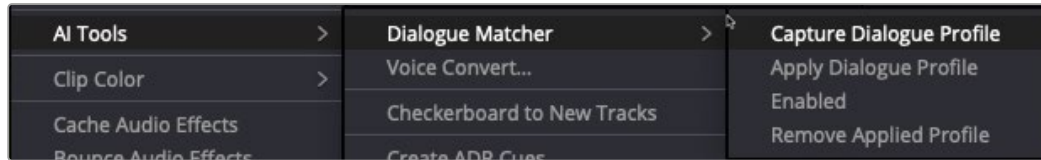


Timeline clip Level Matcher contextual menu

- 2 Right-click on a timeline clip you want to adjust and choose Clip Operations > Level Matching > Apply Level Profile. Repeat this step for any other timeline clips you want to change.

AI Dialogue Matcher

AI Dialogue Matcher is helpful when you need to match the ambient quality of one timeline clip to another. For example, in a dialogue replacement scenario, you may need to make the newly recorded dialogue sound like it was recorded in the same room as the rest of the on-location dialog.



Timeline clip AI Dialog Matcher contextual menu

To apply AI Dialogue Matcher:

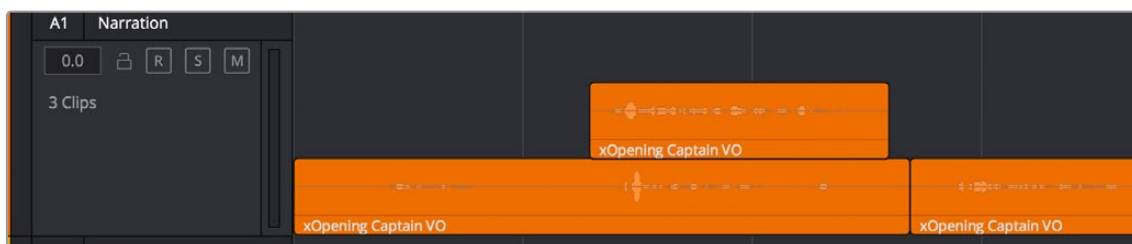
- 1 Right-click the timeline clip you want to match and choose AI Tools > Dialogue Matcher > Capture Dialogue Profile from the contextual menu.
- 2 When the audio has been analyzed, right-click the timeline regions you want to apply the profile to and choose AI Tools > Dialogue Matcher > Apply Dialogue Profile from the contextual menu.

Once the dialogue profile is applied, you will see a checkmark next to the Enabled option in the Dialogue Matcher contextual menu.

- Clicking this checkmark toggles the dialogue profile on and off.
- Selecting Remove Profile removes the profile from the clip entirely.

Audio Clip Layering

Audio layering is a special audio editing mode that lets you superimpose multiple audio clips in the same track, with audio clips edited into the top layers muting overlapping sections of audio clips appearing on lower layers. With audio layering enabled, superimposed audio clips are treated similarly to superimposed video clips that have opacity set to 100%, with clips on top obscuring (or muting) clips underneath.



An example of multiple audio performance editing using layers, where the top layer mutes overlapping sections of audio clips in lower layers

Audio layering is incredibly useful for any situation where you're combining segments of multiple takes together to create a single voiceover, audio vocal track, or dramatic performance, as you can choose which segments to use via their superimposed position in the stack of clips appearing in that track, while at the same time you're preserving the other takes underneath in case you might want them later.

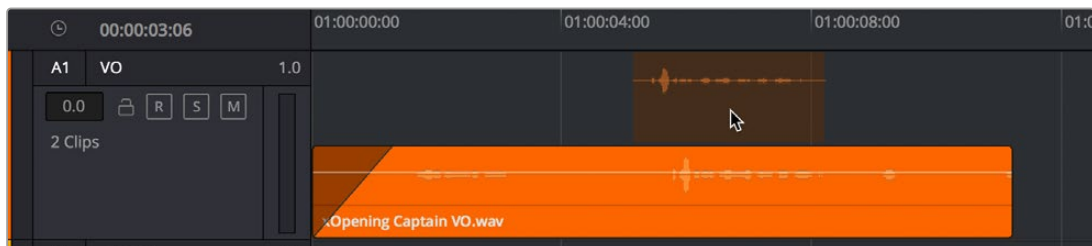
TIP: Track layering can be used on the Edit page as well.

To enable audio layering:

- Choose Timeline > Layered Audio Editing so that a check mark appears by the command. All overlapping audio will be layered instead of overwritten from that point forward.

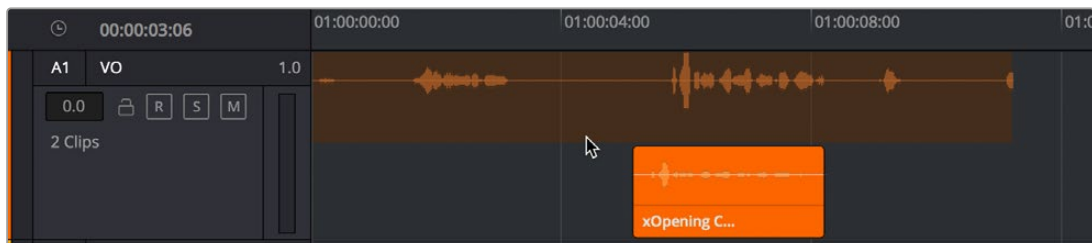
To view audio layering:

- 1 Choose View > Show Audio Track Layers to reveal track layers for each audio track (and each lane within a given audio track) in the Timeline. When layering is on, space appears at the top of each track in the Timeline which provides a region into which you can edit layered audio clips.
- 2 To edit an audio clip or segment as a layer within a particular audio track, drag it from elsewhere in the Timeline or from the Media Pool, and drop it into the empty area above whatever audio is already in that track.



Dragging an audio clip to become layered above another clip

- 3 Edit the different superimposed layers of audio such that the segments of each take that you like are on top. Only the topmost clip segments will be audible. Audio segments that overlap underneath are silent. To put another layer on top, drag it from its current position to the empty area at the top of the track.



Dragging a bottom audio layer to appear on top

- 4 When you're finished editing clips in track layers, choose View > Show Audio Track Layers again to hide the individual layers, so that only the topmost clips appear as a flat sequence in each track and lane.

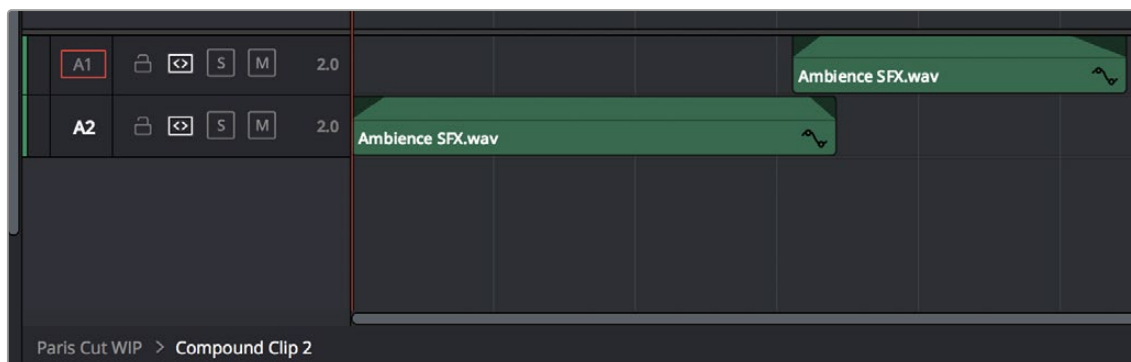


Hiding layers makes all audio layers appear on a single track, as if the top clips are overwriting what's underneath, except the muted material underneath is preserved

Audio Compound Clips

DaVinci Resolve supports audio compound clips, which are created just like any other compound clip, by selecting multiple audio clips, right-clicking one of them, and choosing New Compound Clip. Alternately, compound clips with video clips may now contain multiple audio items as well.

When compound clips containing audio are opened in the Edit or Fairlight pages by right-clicking an audio compound clip and choosing Open in Timeline, breadcrumb controls appear beneath the Timeline that let you exit the compound clip and get back to the master Timeline.

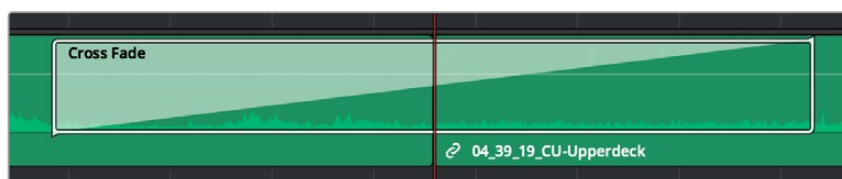


Opening an audio compound clip; note the path control at the bottom left of the Timeline

Audio Crossfades

You can add Cross Fade transitions to any edit point between two audio clips that have enough handles similarly to how you add video transitions, by dragging and dropping from the Effects Library, by right-clicking an edit and choosing an option from the contextual menu, or by selecting an audio edit point and choosing Timeline > Add Audio Only Transition (Shift-T).

Cross Fade transitions are a quick and easy way to fade the volume of the outgoing clip down while simultaneously fading the volume of the incoming clip up, letting you create a smooth aural transition between two audio clips.



An audio Cross Fade transition applied between two clips

You can double-click a Cross Fade transition to open it into the Inspector, revealing the following parameters:

- **Duration:** The duration of the transition, shown in both seconds and frames.
- **Alignment:** A dropdown that lets you choose the transition's position relative to the edit point it's applied to. Your choices are "End on Edit," "Center on Edit," and "Begin on Edit."
- **Transition style:** You can choose -3dB, 0dB, or +3dB to set both the Fade In and Fade Out levels to the same value. For more information on what these levels mean, see the following parameter.
- **Fade In/Fade Out levels:** There are three options that affect the incoming and outgoing halves of the Cross Fade effect independently. 0dB applies a linear fade (this is the default). +3dB applies a

boosted curve; when applied to both Fade In and Fade Out, this can compensate for diminished levels in the middle of a Cross Fade. -3dB applies an attenuating curve, which deliberately lowers the level of the Cross Fade.

Crossfades can be created and edited on both the Edit and Fairlight pages.

Fades and Crossfades

Part of audio editing in the Fairlight page is the use of fades and crossfades. This section shows you how to create these effects for smoothly segueing from one audio clip to another.

Basic fades can be created on Dolby Atmos master files placed on a timeline as a clip. However, complex fades and batch fades are not supported.

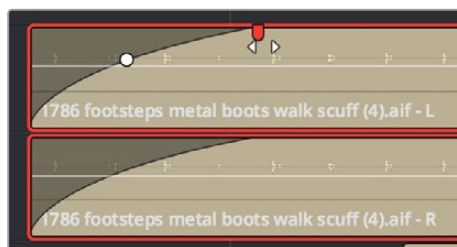
Using Fades

Like the Edit page, each audio clip has fader handles that appear at the upper right and left corners of a clip you're hovering the pointer over.



Fader handles appear when you hover the pointer over a clip

Pulling these handles out creates a fade effect with a duration equal to the length you extended the handle.



Creating a fade effect by pulling out one of the fader handles

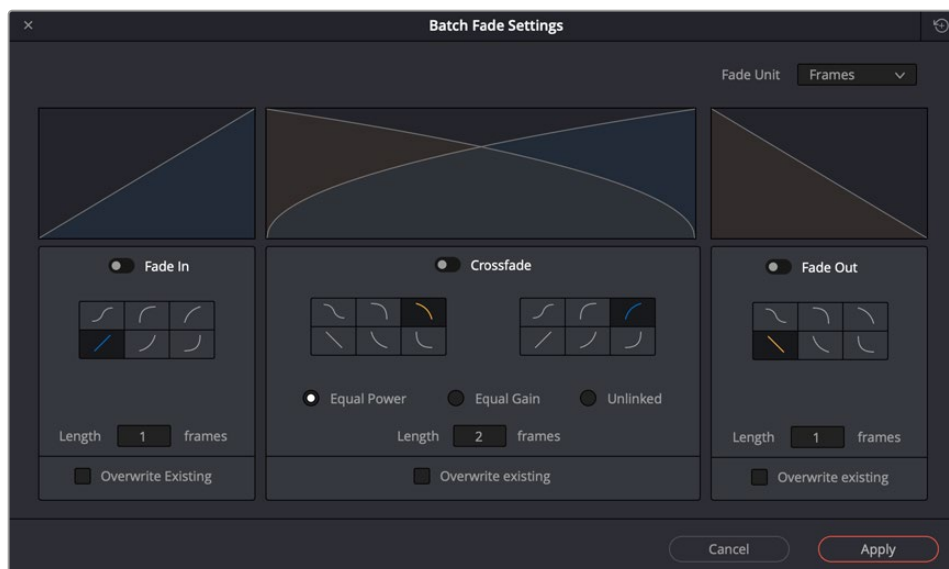
Once you've created a fade effect, you can adjust the curve of the fade by dragging the handle that appears right on top of the fader curve. Dragging the handle up and down affects the angle of the curve, and dragging the handle left and right affects the shape of the curve. In this way, you can create all manner of fade effects.



Adjusting the curve of the fade

Batch Fade and Crossfade Editor in the Fairlight Page

Multiple fades are available to multiple clips with multiple tracks selected, significantly increasing fade functionality. The Batch Fades window has fade shapes for Fade In, Cross Fade, and Fade Out. The Fade Length is user definable by frame for each of these fade types, and there is an option to overwrite existing fades on the highlighted clips.



All of the settings in the Batch Fades window

There are six options per Fade type for precise usage. When adding Fades between sources, it's important to determine the shape so that only the desired media is included in the fade. The curves indicate the ramping slopes of the fade.

Each box has similar controls for affecting the individual Fade type.

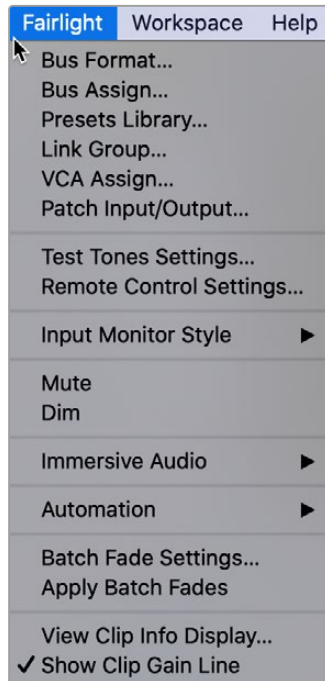
- 1 Next to the Fade type name is the on/off toggle
- 2 There are six Fade curves to create the precise fade needed.
- 3 A Length box determines how many frames the fade will extend,
- 4 A click box to overwrite any previous fades, if needed.

The Crossfade box has three additional controls.

- 1 Equal Power maintains the signal level through the fade.
- 2 Equal Gain is used when crossfading media that could phase when combined. For instance, if a music cue is crossfaded at the mid-point of the fade, there could be a jump in level or added phasing issues. Using this option maintains the gain across the fade for that type of media.

- 3 Unlinked allows for a different curve type on either side of the crossfade.

This window will retain the settings that were last made so once a series of Fade types have been chosen, then Batch Fades can be made without opening this window and can be accessed via the Fairlight menu with a single click across highlighted regions. The Apply Batch Fades in the Fairlight menu are applied with the settings made in the Batch Fades window.



Batch Fade options in the Fairlight menu

This graphic shows highlighted clips and tracks ready to have Batch Fades applied.



Tracks and clips highlighted for Batch Fades

The graphic below shows how the Batch Fades are applied. Clips that are not connected have Fade In and Fade Out applied. Clips that are connected have Crossfades applied. The Batch Fades window determines each of the Fade parameters and then applies them to all of the selected clips. In this example, the Overwrite box was unchecked. Compare the fades that were already there and see that they retain their original Fade length and shape.

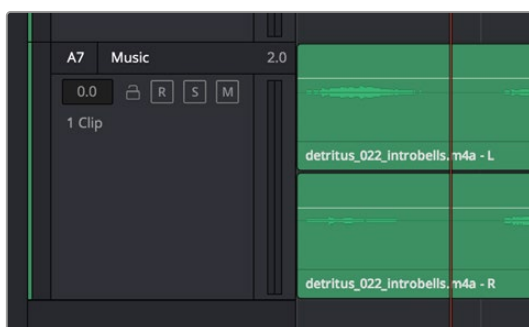


All of the highlighted clips have been faded according to the Batch Fades window.

NOTE: If the media is too short, a dialog box will alert you that there are insufficient handles and that the fade could not be created. You can then choose to trim clips, skip clips, or cancel the Batch Fade. If you cancel, you can then alter the settings to better suit the media.

Fade In and Out to Playhead

A pair of commands in the Trim menu let you move the playhead over a clip, and use the playhead position to “Fade In to Playhead” or “Fade Out to Playhead.” These commands work in both the Edit and Fairlight pages.



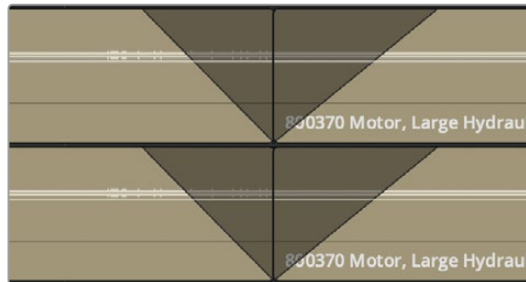
(Left) Placing the playhead where you want a fade in to end, (Right) Using Fade In to Playhead

Creating Crossfades With Overlapping Fades

While a fade gradually fades a single track of audio up or down, a crossfade fades two overlapping clips at the same time, fading one clip up and another clip down, for the aural equivalent of a cross dissolve. There are currently two ways of creating a crossfade in the Fairlight page. Both depend on clip layering to allow you to have overlapping fades over overlapping clips.

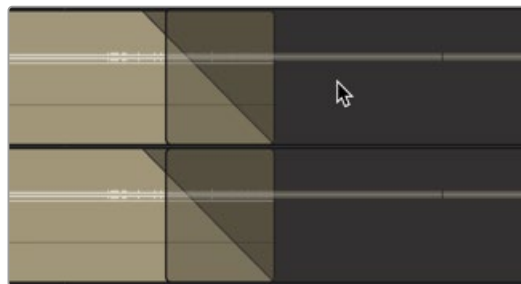
To create a crossfade by overlapping two clips together:

- 1 Add a fade out to the end of one clip, and a fade in to the beginning of another clip. By default, all fades you add in this way are linear, although you can adjust them to whatever gain you want.



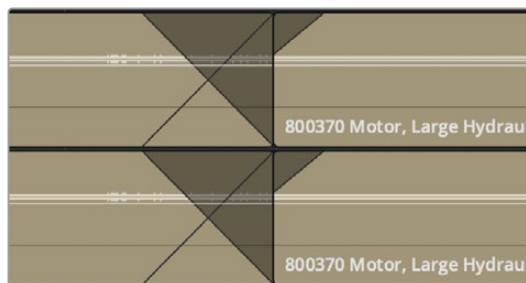
Adding fades to adjacent clips

- 2 Drag the first clip to overlap the second clip by the length of the fade you've created.



Dragging the clips to overlap, the overlapping parts will be preserved via clip layering

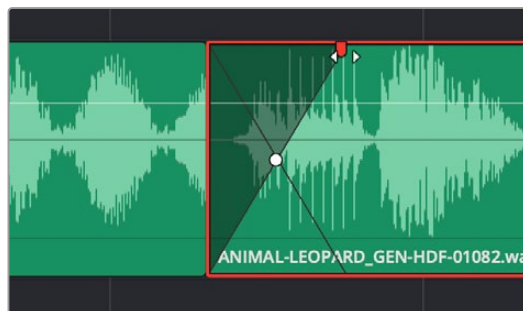
- 3 Drop the clip. The overlapping fades will both be preserved thanks to clip layering, and a crossfade will appear in the Timeline.



The resulting crossfade

To create a crossfade over two clips that are already layered:

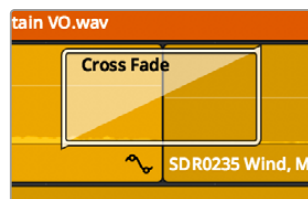
- Drag a fader handle at the beginning or end of a clip that's layered, and a crossfade will automatically appear.



Creating a crossfade by adjusting the fader handle of a layered clip

Using Crossfades From the Edit Page

You can also add crossfades in the Edit page, but they appear in the Fairlight page as transitions in the Edit page style.



The resulting crossfade

Finding Clips in the Media Pool

You can right-click any clip in the Timeline and choose Find in Media Pool to automatically select that clip in the Media Pool, for instances where you might want to edit another copy of that clip somewhere else in the Timeline, or re-edit another segment from that clip into the same area.

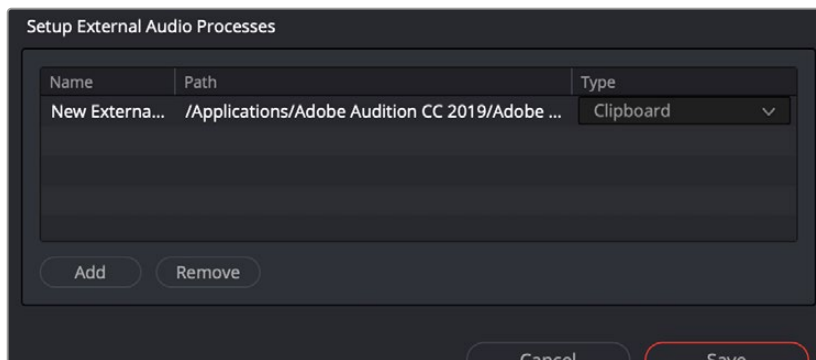
NOTE: Automation Follows Edit when clips with embedded automation are moved or pasted from the Timeline. When clips with automation are instead pulled from the Media Pool, they are in their default state with no automation attached.

Changing Clip Color in the Timeline

You can right-click one or more selected clips in the Timeline to change the clip color, to be more organized. For example, you might set production audio clips containing dialog from different actors to different colors, or you could set clips with dialog, music, and effects to different colors in order to easily differentiate each clip's purpose.

Editing Audio Clips in External Editors

While working in the Fairlight page, you have the ability to process an audio file using a third-party application if necessary, in the event you need to use another application's capabilities to create an effect or solve an issue that can't be accomplished in the Fairlight page itself. To do this, you must first add one or more applications to the External Audio Process list in the Audio Plugins panel of the System Preferences.



The External Audio Process list configured to send audio to Adobe Audition

To add an external audio process:

- 1 Open the System tab of the DaVinci Preferences, and select the Audio Plugins panel.
- 2 Click the Add button in the “Setup External Audio Processes” section.
- 3 To give the audio process a different name, double click in the Name column and type a new name.
- 4 Double-click in the empty Path column for the new process, and choose an application to assign to that process from the dialog.
- 5 Choose the type of process you want it to be from the dropdown menu in the Type column.

Once you have one or more external audio applications configured in Preferences, you can use them to process any audio clip in the Fairlight page by right-clicking an audio clip and choosing the application you want to use from the External Audio Process submenu of the contextual menu.

When you do this, a duplicate of the audio clip media is copied (bounced) to the directory location specified by the “Save clips to” field of the Capture and Playback panel of the Project Settings. At that point, the external application is either opened or launched as a command from the command line (depending on how the external application has been configured in Preferences).

Once the bounced audio is opened in the external application, you can process it however you need to and bake in any changes made by saving/rendering/outputting and overwriting the original copied audio media file. DaVinci Resolve detects when changes have been made, and the altered result is automatically reimported as an additional audio layer on top of the original clip in the Timeline.

The way an audio application is configured in the DaVinci System Preferences dictates how the bounced audio file is passed to the external program. There are three choices:

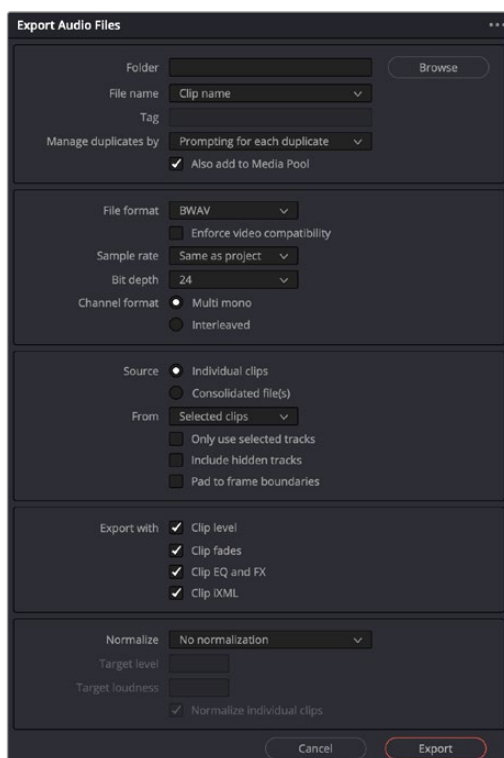
- **Command Line:** As a command line parameter, if your audio application is able to be run from the Terminal.
- **Clipboard:** By placing the path to the bounced file in the clipboard, so you can paste it into the application which has been automatically launched, or import it via a File > Open dialog.
- **Reveal:** By revealing the bounced copy in the file manager of your workstation, so you can drag and drop it onto the application which has been automatically launched.

Exporting Audio Clips to External Files

For any workflow where you need to write clips from the Timeline to external files with extensive file altering capabilities, you can use the Export Audio Files command.

To export clips to external files:

- 1 Select one or more clips in the Timeline.
- 2 Right-click the desired selected files and choose Export Audio Files from the contextual menu.
- 3 When the Export Audio Files dialog appears, choose the following:
 - a) Click Browse to choose a location to save the exported audio files.
 - b) (Optional) Enter a tag and one of the various Name options.
 - c) Choose a File Format, Sample Rate, and Bit Depth for the exported files.
 - d) Choose a Channel Format of Multi-Mono or Interleaved for the exported files.
 - e) Choose a Source of Individual Clips or Consolidated Clips; a dropdown menu further defines the ranges of the complete timeline, a selected range, or just the selected clips.
 - f) Checkboxes can further define using only the selected tracks, including hidden tracks and padding to frame boundaries.
 - g) When exporting you can choose to include the Clip Level, the Clip Fades, the Clip EQ and FX, and the Clip iXML.
 - h) The exported files can be normalized to predetermined target levels and target loudness values with all of the standards offered throughout the Fairlight page.



The dialog that lets you choose options for exporting clips to files

- 4 Click Export.

The selected audio files are written to the location you chose.

Sample Editing

You can zoom quite far into audio clips on the Fairlight page timeline, until you see the individual samples that comprise the audio waveform of each clip. Samples are represented by control points once you've zoomed in far enough.



When you zoom in far enough, you can see the individual samples of an audio clip as control points

You can non-destructively edit these control points to eliminate clicks and pops, and to effect other fixes to problem audio clips.

Methods of editing audio samples:

- **To see the editable audio samples:** Zoom all the way into an audio clip until you see the sample control points, using either Command-Plus or Command-Minus, the scroll wheel of your pointing device, or by holding down the ZOOM button of your Fairlight editing panel and turning the JOG/EDITING wheel.
- **To edit a single audio sample:** Click and drag that audio sample up or down to change its height.
- **To edit a section of samples:** Click and drag horizontally left or right across the samples you want to edit to “redraw” the waveform any way you’d like.
- **To reset all edited samples to their original state:** Right-click an audio clip with edited samples, and choose Reset Edited Samples from the contextual menu.



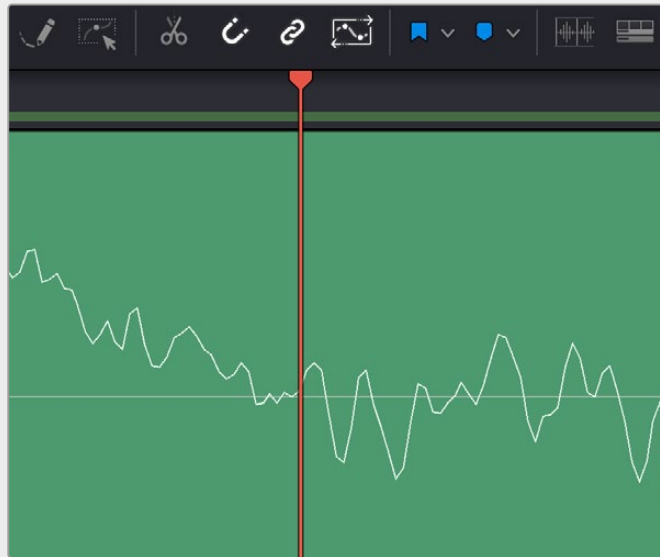
Black points show the previous levels when samples have been edited in a clip

Sample editing can be undone, just like any other editing procedure, as the edited sample points are stored non-destructively within the DaVinci Resolve project.

Waveform Zero Crossing Indicator

Waveforms have a zero crossing indicator line. Since a waveform is an image representation of sound continually moving positive to negative, the zero crossing point is the level at which that fluctuation occurs. When zooming into a waveform at the sample level, the waveform will display the zero crossing line to enable precise editing.

The zero crossing is a useful feature when editing audio, since clean edits are made at the zero crossing to avoid inducing clicks or pops. A crossfade between two audio clips automatically brings both sides of the fade to the zero crossing.



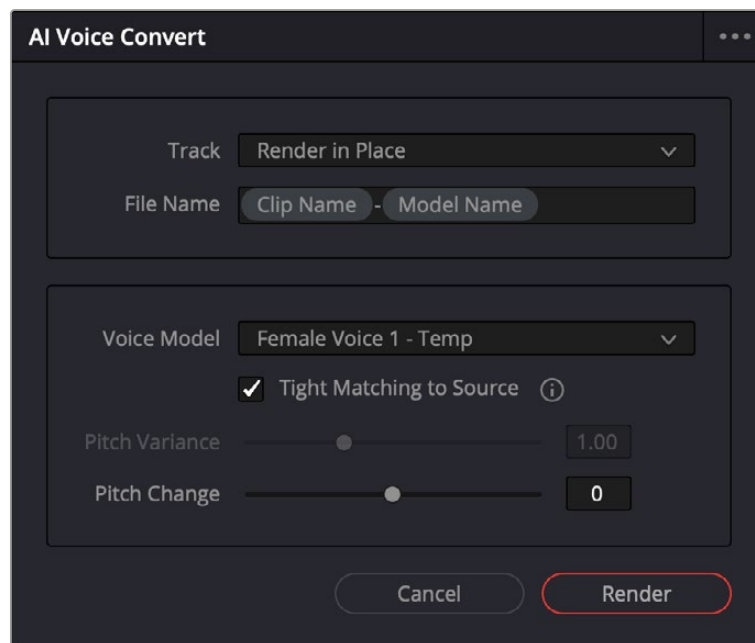
The center line in the waveform indicator is the zero crossing.

Voice Convert (Studio Version Only)

DaVinci Resolve has an incredibly sophisticated AI-driven voice conversion tool, that allows you to essentially swap one speaking voice for another. The use cases for this tool are wide reaching and even somewhat controversial, however this Voice Convert toolset can be an invaluable resource in a post production. It allows you to take a generated voice model that has been captured by DaVinci Resolve and “drive” that voice model using another voice tracks recording as its foundation.

In simple cases, you can replace your own voiceover tracks with a voice that has more depth and gravitas to it, or even switch the gender of your own voice if you feel it's more appropriate for the subject matter. Perhaps you recorded audio in a noisy environment and wanted to make a pristine version instead by using your own voice as a model, to replace your original noisy tracks. In more complicated scenarios you can train your own voice convert model on your actor's voice patterns and use that to change and replace their previous voiceover, without having to get them back into the recording booth. Finally, it is even possible to change the voice of an actor completely to another voice, all while staying perfectly in sync with the picture. The inflection, pitch variations, and emotion of the original all remain; it just sounds like someone else.

Needless to say, using this powerful tool requires a certain level of responsibility on the part of the user. Privacy, copyright, and union laws and rules are all in flux at the moment around AI-generated content, and it's necessary to carefully navigate the ethics involved in its professional use.



The Voice Convert tool

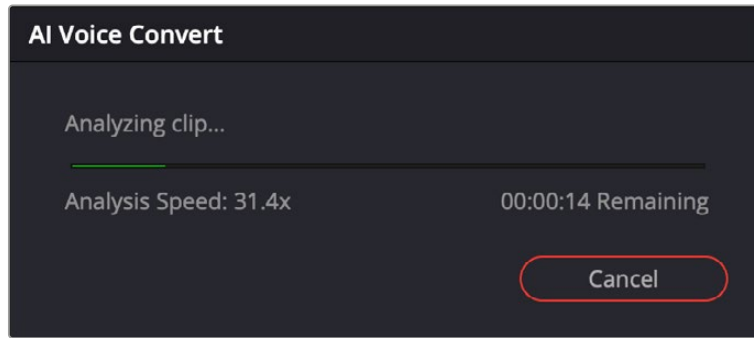
NOTE: Using Voice Convert is very processor intensive and requires your system to have a minimum of 8GB of video RAM before it will activate to avoid crashes and system instability. Even with a strong system, creating a Voice Model will require prolonged intensive CPU and GPU power, so multi-tasking while it's working may be challenging.

Using Voice Convert

Like most of DaVinci Resolve's AI tools, using Voice Convert is very simple, with all of the complex processing involved handled by the voice model.

To use Voice Convert to change one voice to another:

- 1 Place the clip or clips you want to change the voice of in the timeline, and select them.
- 2 Right-click on the clip and select Voice Convert from the context menu, or select Clip > AI Tools > Voice Convert in the main menu.
- 3 Select the track that you want the new audio to appear in (including overwriting the original).
- 4 Select the file name you want the new audio to be named.
- 5 Select the Voice Model you want to use for the new audio (you can make your own custom voice models as explained below), and its parameters.
- 6 Click Render to start the Voice Convert process.



Voice Convert analyzing a clip

At this point the Voice Convert dialog will appear showing you the progress of the conversion. Depending on the length and number of clips selected, this can take some time. When the process is done, the new audio will appear in the timeline in the exact same place as the original audio (or on the same place in another track if that option was selected). The new audio clips will also appear in the Media Pool.

Simply play back the timeline to audition the new voice. You can change the audio back to the original by right-clicking on a clip and selecting Revert to Original Voice from the context menu.

TIP: While you can drive voices with different accents, Voice Convert generally can't recreate another accent. For example, if you have a voice model that is a speaker with a British accent and drive that voice with an American accent, you get more of an American version of the British accented speaker. But if you speak with a British accent when driving the British voice model, it will work well.

Voice Convert Settings

Each voice conversion requires selecting set up parameters to drive the model.

- **Track:** Select which track you want the new audio to be placed on. The default is Render in Place, which will overwrite the original audio.
- **File Name:** Set the file name for the new audio clip. You can enter a specific name, or use the metadata variables to create a template for file naming. By default the name is ClipName-(Voice Model Name).
- **Voice Model:** Select which voice model you wish to use for the new audio. There are some built-in options that come with DaVinci Resolve, and any custom options you create will also be available here.
- **Tight Matching to Source:** Check this box to tightly follow the pitch, intonation, and volume variance of the source voice. Uncheck this box to have a looser interpretation using the Pitch Variance tool.
- **Pitch Variance:** When the Tight Matching to Source box is unchecked, this tool becomes active. Lower values create a more monotone interpretation, while higher values have more variance in pitch.
- **Pitch Change:** Use this slider to create a deeper (negative values) or higher (positive values) voice based on the model. The values are in semitones and can be adjusted in 100 cents (.1 semitone) increments.

Creating Your Own Voice Models

It is possible to create your own voice models for use with Voice Convert. These can be of your own voice or even someone else's that you have the rights to. This section goes over some best practices to get a clean and usable voice model.

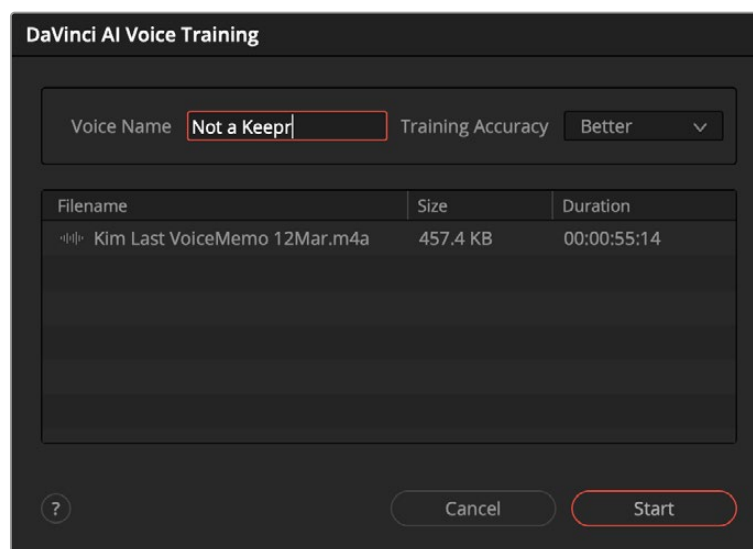
Preparing the Voice Model Assets

You will get the best results using ten minutes or so of clean, high-quality recordings of the voice to be modeled. The audio should be free of excessive ambiance and noise, with little to no dynamics processing. You will want consistent material that sounds natural. It can be useful to have a variety of emotions expressed as well, if you want to hear those results for the driven voice. Treat it like a voice over audition; show your range.

These voice assets do not necessarily need to be in one long clip, though if there are a lot of separate files, it can be helpful to consolidate them all in a single bin for organization. The multiple clips should be of consistent quality, without audio variations in the recorded sources.

Creating the Voice Model

To generate your own voice model, select one or more clips for analysis in the Media Pool, right-click and choose AI Tools > DaVinci AI Tools Voice Training from the context menu. If this is the first time you've run this tool, DaVinci Resolve will open up the Resolve Extras Download Manager to install the required support files.

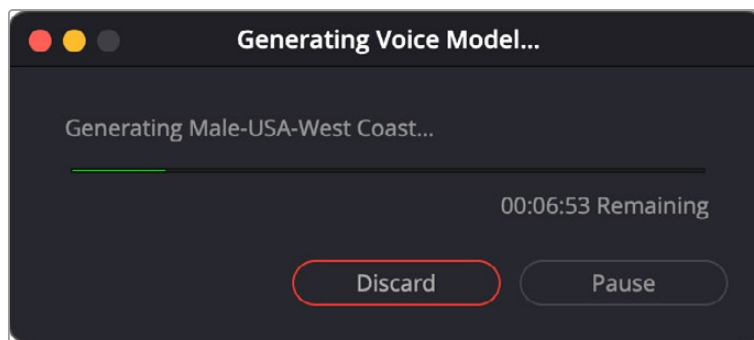


Voice Model generation

Initially, a warning splash screen appears to ensure you assume full responsibility for how you will use this tool. Once you hit accept, The DaVinci AI Voice Training dialog appears, and there are two options to set.

- **Voice Model Name:** Enter a descriptive name for your new model.
- **Quality:** Choose Faster or Better mode for analysis. Better takes about 3x the time as faster, and 10 minutes or so of material can take a few hours to complete, depending on your computer's power.

Initially creating a voice model launches a foreground process, then once the initial analysis is complete, it will switch to a background process, allowing you to continue to use DaVinci Resolve for other purposes or other apps while it finishes.



Voice Model Progress

You can see the progress of the model in the Voice Model Generation Progress indicator in the bottom right corner, or by selecting **Workspace > Voice Model Progress**.



Voice Model Generation Progress indicator will slowly fill blue as the voice model finished. Hover over the top to get an ETA and progress finished tooltip.

You can also alternatively choose to pause or discard the in-progress model instead. Quitting DaVinci Resolve will also pause the background process, and it will automatically start up again the next time DaVinci Resolve is launched.

Once the model is complete, the new voice preset will appear for use in the Voice Convert dialog.